

CS601 DATA
COMMUNICATION

UPDATE NOTES
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Data-Link Layer: Data Link layer controls node-to-node communication.

Nodes and Links: We refer to the two end hosts and the routers as nodes and the networks in between as links.

Services provided by Data-Link Layer

- Located between the physical and the network layers
- Provides services to Network Layer and receives services from Physical layer
- Framing
- Flow Control
- Error Control
- Congestion Control

Two Categories of Links

- Two nodes are physically connected by a transmission medium such as **cable or air**
- Data-link layer controls **how the medium is used**
 - ☐ Data-link layer can use whole capacity
 - ☐ Data-link layer can use only part of the capacity
- We can have the following **two types of links:**
 - Point-to-point link or a
 - Broadcast link

Two Sublayers of Data-Link Layer

- We can divide the data-link layer into two sublayers:
- Data Link Control (DLC) (It **broadcast links** and it provides **point to point links**)
- Media Access Control (MAC) (It only **broadcast links**)

Why Link Layer Addressing?

- IP addresses are the identifiers at the network layer
- In Internet we cannot make a packet reach its destination using only IP addresses
- Source and destination IP addresses define the two ends but cannot define which links the packet will take

Three Types of Addresses

- Unicast
- Multicast
- Broadcast

Example of Unicast address: The unicast link-layer addresses in the most common LAN, Ethernet, are

48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons; for example, the

following is a link-layer address of a computer. **The second digit needs to be an odd number.**

A3:34:45:11:92:F1

Example of Multicast address: The multicast link-layer addresses in the most common LAN, Ethernet,

are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons. **The second digit,**

however, needs to be an even number in hexadecimal. The following shows a multicast address:

A2:34:45:11:92:F1

Example of Broadcast address: The broadcast link-layer addresses in the most common LAN, Ethernet,

are 48 bits, all 1s, that are presented as 12 hexadecimal digits separated by colons. The following shows a

broadcast address: **FF:FF:FF:FF:FF:FF**

Address Resolution Protocol:

- Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node

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- IP address of the next node is not helpful in moving a frame through a link; we need the link layer address of the next node

Data can be corrupted during transmission. Some applications require that errors be corrected.

Types of Errors:

In a **single-bit error**, only one bit in the data unit has changed.

A **burst error** means that two or more bits in the data unit have changed.

Redundancy:

Central concept in detecting or correcting errors. We send **extra (redundant) bits** with data.

- **Detection vs Correction:**

Correction is more difficult than the detection

In **error detection**, we are only looking to see if any error has occurred (Yes)

- We are not interested in the number of corrupted bits in Detection
- Single-bit error is same as a Burst error

In **Error Correction**, we need to know:

- The exact number of bits that are corrupted and,
- Their location in the message

There are two main methods of error correction: forward error correction and correction by retransmission.

Coding:

In coding, we need to use modulo-2 arithmetic. Operations in this arithmetic are very subtraction give the same results. We subtraction.

Types of coding schemes:

We can divide coding schemes into two broad categories: block

Block coding: In block coding, we divide our message into blocks, each of k bits add r redundant bits to each block to make the length **codewords**.

Error detection in block coding: In block coding, errors be detected by using the following two conditions:

1. The receiver has (or can find) a list of
2. The original codeword has changed to an invalid one.

Example: Let us assume that $k = 2$ and $n = 3$. Table below shows the list of datawords and codewords.

Later, we will see how to derive a cod

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detected and

, e.g., 1 to 0 **or** 0 to 1

e.g., 1 to 0 **and**

correcting errors is Redundancy. To detect or correct errors, we need to

, Yes or No)

, n simple; addition and

use the XOR (exclusiveOR) operation for both addition and

coding and convolution coding.

bits, called

$n = k + r$. The resulting **n -bit blocks**

valid codewords.

code word from a dataword.

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linkdetected

0 to 1.

, **datawords**. We

are called

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The Hamming distance: The Hamming distance between two words is the number of differences

between corresponding bits. The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.

Minimum Hamming distance: To guarantee the detection of up to s errors in all cases, the minimum

Hamming distance in a block code must be

cases, the minimum Hamming distance in a block code must be

Example: A code scheme has a Hamming distance

how many errors?

$d_{min} = 4$, $d_{min} = s + 1$, $S = d_{min} -$

Block code guarantees detection of

A linear block code: In a linear block code, the exclusive OR (XOR) of any two valid codewords creates

another valid codeword.

Example: The code below is a linear block code because the result of XORing any codeword with any

other codeword is a valid codeword. For example, the XORing the fourth one

.

The numbers of 1s in the nonzero codewords are 2, 2, and 2. So the minimum Hamming distance is d_{min}

= 2.

Parity-check code: A simple parity

$d_{min} = 2$. A simple parity-check code can detect an odd number of errors. All Hamming codes

disc

in this book have $d_{min} = 3$. The relationship between

Cyclic codes: Cyclic codes are special linear b

codeword is cyclically shifted (rotated), the result is another codeword.

cyclically left-shift, then 0110001 is also a codeword

Advantages of Cyclic codes:

- Good performance in detection:
- Single-bit errors
- Double errors
- Odd number of errors
- Burst errors
- Easy Implementation
- Fast Implementation

Cyclic redundancy check (CRC) :

is used in networks such as LANs and WANs.

Checksum: Error-detection technique that can be applied to a message of any length

used at the network and transport layer rather than the data link layer using a 16-bit checksum, which uses one's complement arithmetic. In this arithmetic, we can represent unsigned numbers between 0 and $2^n - 1$.

Example: Suppose the message is a list of five 4-bit numbers. In addition to sending these numbers, we send the sum of the numbers. Set of numbers is (7, 11, 12, 0, 6). Sum=36

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Hamming distance = $s + 1$. To guarantee correction of up to t errors in a code of length n , $d_{min} = 2t + 1$.

Hamming code = 4. This code guarantees the detection of up to 3 errors, $S = 4 - 1 = 3$.

Parity bit = up to 3 bits in a codeword.

The first bit of the second and third codewords creates

parity-check code is a single-bit error-detecting code in which

m and n in these codes is $n \geq 2m$.

Block codes with one extra property. In a cyclic code,

If 1011000 is a codeword and we

perform a left shift, the result is also a codeword. A category of cyclic codes called the cyclic redundancy check (CRC)

is used for error detection. Checksum mostly

operates at the data-link layer. Traditionally, the Internet has been

using only $n - 1$ bits.

For 4-bit numbers that we want to send to a destination. In

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distance between

. all

which $n = k + 1$ with

discussed

1.

lock if a

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Data link control (DLC): Data link control deals with the design and procedures for communication between two adjacent nodes: node-to-node communication. Data link control functions include

framing, **flow control** and **error control**.

Framing in the data link layer separates a message from one source to a destination, or from

other sources to other destinations. Frames can be of **fixed or variable size**.

In fixed-size framing, there is no need for defining the boundaries of frames, size acts as a

boundary/delimiter; **in variable-size framing**, we need a delimiter (flag) to define the boundary of two

frames. We have to define beginning and ending of frame.

Variable-size framing Techniques:

Variable-size framing uses two categories of protocols: **byte-oriented (or character-oriented) and bitoriented**.

☐ **In a byte-oriented protocol**, the data section of a frame is a sequence of bytes; **in a bitoriented**

protocol, the data section of a frame is a sequence of bits.

☐ **In byte-oriented (or character-oriented) protocols**, In order to avoid confusing the receiver we use byte stuffing; a special byte added to the data section of the frame when there is a character with the same pattern as the flag.

☐ **In bit-oriented protocols**, we use bit stuffing; an extra 0 is added to the data section of the frame when there is a sequence of bits with the same pattern as the flag.

Flow control and error control:

Flow control: Flow control refers to a set of procedures used to restrict the amount of data that the sender

can send before waiting for acknowledgment.

Error control: Error control refers to methods of error detection and correction.

Connectionless or connection-oriented protocol:

A DLC protocol can be either connectionless or connection-oriented

☐ Connectionless: No relationship between the frames

☐ Connection-Oriented: Frames are numbered and sent in order

Finite State Machine (FSM):

☐ A machine with a finite number of states

☐ Machines stays in one of the states until an event occurs

☐ Each event is associated with 2 reactions:

☐ List of actions to be performed

☐ Determining the next state

For the noiseless channel, we discussed two protocols: the **Simplest Protocol** and the **Stop-and-Wait**

Protocol.

The first protocol has neither flow nor error control; **the second** has no error control.

☐ **In the Simplest Protocol**, the sender sends its frames one after another with no regards to the receiver.

☐ **In the Stop-and-Wait Protocol**, the sender sends one frame, stops until it receives confirmation from

the receiver, and then sends the next frame.

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Piggybacking: Both Simple and Stop

A technique called **piggybacking** is

frame is carrying data from A to B, it can also carry control

frame is carrying data from B to A, it can

High-level Data Link Control (HDLC):

protocol for communication over point-to-point protocol.

HDLC transfer modes:

HDLC provides two common transfer modes that can

- ☐ Normal Response Mode (NRM) &
- ☐ Asynchronous Balanced Mode (ABM)

Framing:

• HDLC defines three types of frames:

- ☐ information frames (I)
- ☐ Supervisory frames (S)
- ☐ Unnumbered frames (U)

Point-to-point protocols (PPP): The Point-to-Point Protocol (PPP), which is a byte-oriented protocol.

Multiplexing in PPP:

- Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate and carry the network-layer data.
- Three sets of protocols are:
 - Link Control Protocol (LCP)
 - Two Authentication Protocols (APs)
 - Several Network Control Protocols (NCPs)

Authentication protocols in PPP:

Media Access Control (MAC) Sublayer:

- When nodes use a multipoint or broadcast link, we need a multiple access protocol to the link.
- Many protocols have been devised to handle access to a shared link.
- All of these protocols belong to Media Access Control (MAC) sublayer.

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Stop-and-wait protocols are designed for unidirectional communication used to improve the efficiency of the bidirectional protocols.

Information about frames from B; when a

frame is received, it also carries control information about frames from A.

High-level Data Link Control (HDLC) is a bit-oriented protocol for point-to-point and multipoint links. It implements Stop-and-wait and sliding window protocols. It can be used in different configurations:

I-frames)

S-frames)

U-frames)

One of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP), a byte-oriented protocol.

It is used to encapsulate network-layer data.

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– **Layer:**

multiple-access protocol to coordinate
sub-layer

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communication.

iciency protocols. When a

bit-oriented

Stop-and-Wait

Point-to-layer

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Taxonomy of Multiple Access Protocols

☐ **Random Access**

☐ In random-access or contention no station is superior to the other and none is assigned control

over the other

☐ Station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send

☐ This decision depends on the state of the medium (idle or busy)

ALOHA

- ALOHA, the earliest random access method, was developed in early 1970s
- Designed for a radio (wireless) LAN, but it can be used on any shared medium
- Potential collisions in this arrangement as the medium is shared between the stations
- When a station sends data, another station may attempt to do so at the same time
- The data from the two stations collide and become garbled

Slotted ALOHA

- We divide time into slots of T_{fr} sec and force the station to send only at the beginning of the slot
- Invented to improve the efficiency of pure ALOHA
- If a station misses the time slot, it must wait until beginning of next time slot reducing vulnerable time to T_{fr} (vs. $2 \times T_{fr}$ for pure ALOHA)

Carrier Sense Multiple Access (CSMA):

- To minimize the chance of collision and, therefore, increase the performance, CSMA was developed
- The chance of collision is reduced as the station is required to sense/listen to the medium before sending data
- 'sense before transmit' or 'listen before talk'

Carrier Sense Multiple Access/ Collision Detection:

- CSMA method does not specify the procedure following a collision
- CSMA/CD augments the algorithm to handle the collision
- The station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision, the frame is sent again

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Carrier Sense Multiple Access/ Collision Avoidance:

- CSMA/CA was invented for Wireless Networks
- Collisions are avoided through the use of three strategies:

- ☐ The Interframe Space
- ☐ The Contention Window
- ☐ Acknowledgements

CSMA/CA

- Interframe Space (IFS): Collisions are avoided by deferring transmission even if the channel is idle
- Contention Window: Amount of time divided into slots. Station chooses a random number of slots as its wait time (one slot first time and double each time system cannot detect an idle channel)
- Acknowledgement: Positive acknowledgement and time-out timer can help guarantee that the receiver has received the frame

☐ **CONTROLLED ACCESS:**

- The stations consult one another to find which station has the right to send
- A station cannot send unless authorized by other stations
- We discuss three controlled-access methods:

- ☐ Reservation
- ☐ Polling
- ☐ Token Passing

☐ **Reservation:**

- In the reservation method, a station needs to make a reservation before sending data
- Time is divided into intervals
- In each interval, a reservation frame precedes the data frames sent in that interval

☐ **Polling:**

- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations
- All data exchanges must be made through primary device even when the ultimate destination is a secondary device

☐ **Token Passing:**

- In the token-passing method, the stations in a network are organized in a logical ring
- For each station, there is a predecessor and a successor
- The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring
- Special packet called TOKEN circulates through the ring
- Possession of TOKEN gives the station the right to send the data
- TOKEN Management is required to manage possession time, Token monitoring, priority assignment etc.

☐ **CHANNELIZATION (Channel Partition):**

- The available bandwidth of a link is shared in time, frequency, or through code, among different stations
- We discuss three protocols:

- ❑ Frequency Division Multiple Access (FDMA)
- ❑ Time Division multiple Access (TDMA)
- ❑ Code Division Multiple Access (CDMA)

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❑ Frequency-Division Multiple Access (FDMA):

- In FDMA, the available bandwidth is divided into frequency bands
- Each station is allocated a band to send its data i.e. each band is reserved for a specific station, and it belongs to the station all the time
- Each station also uses a bandpass filter to confine the transmitter frequencies

❑ TDMA

- Stations share the bandwidth of the channel in time
- Each station is allocated a time slot during which it can send data
- Each station transmits its data in its assigned time slot

❑ Code Division Multiple Access (CDMA)

- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link
- CDMA differs from TDMA in that all stations can send data simultaneously; there is no timesharing

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Ethernet Protocol:

Ethernet is the most widely used local area network protocol

The IEEE Project 802:

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards

to enable inter-communication among equipment from a variety of manufacturers

- Project 802 did not seek to replace any part of the OSI model or TCP/IP protocol suite

ETHERNET Evolution:

- The Ethernet LAN was developed in the 1970s
- Since then, it has gone through four generations:

- ❑ Standard Ethernet (10 Mbps)
- ❑ Fast Ethernet (100 Mbps)
- ❑ Gigabit Ethernet (1 Gbps)
- ❑ 10 Gigabit Ethernet (10 Gbps)

❑ Standard Ethernet (10 Mbps):

- The original Ethernet technology with the data rate of 10 Mbps is called Standard Ethernet
- It is connectionless and unreliable service
 - o The IEEE 802.3 Standard defines I-persistent CSMA/CD as the access method for first-generation 10-Mbps Ethernet.

ADDRESSING in Standard Ethernet:

- Each station on Ethernet has its own network interface card (NIC)
- The NIC fits inside the station and provides the station with a link-layer/physical address

- The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between bytes.

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- For example, the following shows an Ethernet MAC address:
4A:30:10:21:10:1A

Example:

Define the type of the following destination address

- 4A:30:10:21:10:1A
- 47:20:1B:2E:08:EE
- FF:FF:FF:FF:FF:FF

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even,

the address is unicast. If it is odd, the address is multicast. If all digits are

Therefore, we have the following:

- This is a unicast address because A in binary is 1010 (even).
- This is a multicast address because 7 in binary is 0111 (odd).
- This is a broadcast address because all digits are Fs in hexadec

Access Method in Standard Ethernet

- Since the network that uses the standard Ethernet protocol is a broadcast network, we need to use an access method to control access to the sharing medium
- The standard Ethernet chose CSMA/CD with 1

Efficiency of Standard Ethernet:

- The ratio of the time used by a station to send data to the time the medium is occupied by this station
- The practical efficiency of standard Ethernet has been measured to be:
- Efficiency = $1/(1 + 6.4 \times a)$

Example: In the Standard Ethernet with the transmission rate of 10 Mbps, we assume that the length of the medium is 2500 m and the size of the frame is 512 bits. The propagation speed of a signal in a cable is normally 2×10^8 m/s.

Solution:

Implementation of Standard Ether

- The Standard Ethernet defined several implementations, but only four of them became popular during the 1980s

Summary of Standard Ethernet Imp

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addresses:

Fs, the address is broadcast.

hexadecimal.

Ethernet:

1-Persistent Method

where a = number of frames that can fit on a medium

ernet:

mplementation:

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Changes in Standard:

- The changes that occurred to the 10 the Ethernet to become compatible with other high

- ☐ Bridged Ethernet
- ☐ Switched Ethernet
- ☐ Full-Duplex Ethernet

☐ **Fast Ethernet:**

- In the 1990s, Ethernet made a big jump by increasing the transmission rate to 100 Mbps, and the new generation was called the Fast Ethernet

- To make it compatible with the Standard Ethernet, the MAC sublayer was left unchanged
- But the features of the Standard Ethernet that depend on the transmission rate, had to be changed

- Goals of Fast Ethernet:

- ☐ Upgrade data rate to 100Mbps
- ☐ Make it compatible with Standard Ethernet
- ☐ Keep same 48-bit address
- ☐ Keep same frame format

Summary of Fast Ethernet Impleme

☐ **Gigabit Ethernet:**

- Need for an even higher data rate resulted in the design of IEEE Standard 802.3z Gigabit Ethernet

Protocol (1000 Mbps)

- The goals of the Gigabit Ethernet were:

- ☐ Upgrade the data rate to 1 Gbps
- ☐ Make it compatible with standard or Fast Ethernet
- ☐ Use same 48 bit address
- ☐ Use the same frame format
- ☐ Keep same minimum and maximum frame lengths

Summary of Gigabit Ethernet Imp

☐ **10-Gigabit Ethernet:**

- The idea is to extend the technology, the data rate, and the coverage distance so that the Ethernet

can be used in LANs and MANs (metropolitan area network)

- The IEEE committee created 10 Gigabit Ethernet and called it Standard 802.3ae

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10-Mbps Standard Ethernet opened the road to the evolution of
t high-data-rate LANs

o **mentation:**

mplementation:

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- 10 Gigabit Ethernet operates only in full-duplex mode, which means there is no need for contention; CSMA/CD is not used in 10 Gigabit Ethernet

- **Four implementations are most common:**

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☐ **Other Wired Networks:**

- **Access Networks**

☐ Networks that connect a small LAN to an ISP

- **Wide Area Networks**

☐ Wired networks used to transfer data over long distances

Telephone Network:

- The telephone network had its beginnings in the late 1800s

- Plain Old Telephone System (POTS) was originally an analog system using analog signals to transmit voice

- With the advent of the computer era, the network, in the 1980s, began to carry data in addition to voice

Major Components:

- The telephone network is made of three major components:

☐ Local Loops

☐ Trunks

☐ Switching offices

- The telephone network has several levels of switching offices:

☐ End offices

☐ Tandem offices

☐ Regional offices

Local-Access Transport Areas (LATAs) :

- A LATA can be a small or large metropolitan area

- A small state may have a single LATA; a large state may have several LATAs

- A LATA boundary may overlap with state boundary; part of a LATA can be in one state, part in another state

Intra-LATA and Inter-LATA Services:

- Services offered by Telephone companies inside a LATA are called Intra-LATA services and between LATAs are called Inter-LATA services.

- Carrier that handles Intra-LATA are called a Local Exchange Carrier (LEC) and the ones that handle Inter-LATA are called Interexchange Carriers (IXCs)

Signaling:

- The telephone network in the beginning, used a circuit-switched network with dedicated links to transfer voice communication

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- The operator connected the two parties by using a wire with two plugs inserted into the corresponding two jacks
- Later, the signaling system became automatic

Services:

- Telephone companies provide two types of services:

- ☐ Analog Services
- ☐ Analog Switched Services
- ☐ Analog Leased Services
- ☐ Digital Services
- ☐ Switched /56 Service
- ☐ Digital Data Service

Digital Subscriber Line (DSL):

- After traditional dial-up modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet
- DSL supports high-speed digital communication over the existing telephone
- DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL)

Cable Network:

- The Cable TV networks were initially created to provide remote subscribers access to TV programs
- Cable networks enabled access to remote broadcasting stations via microwave connections
- Cable TV also found a good ISP market by using some of the channels originally designed for video
- Cable TV started to distribute broadcast video signals to locations with poor or no reception in the late 1940s
- It was called community antenna television (CATV) because an antenna at the top of a tall hill or building received the signals from the TV stations

Hybrid Fiber Coaxial (HFC) Network:

- Second generation of cable network is called a Hybrid Fiber-Coaxial (HFC) network
- The network uses a combination of fiber-optic and coaxial cable

Cable TV for Data Transfer:

- Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer

- DSL technology provides high-data-rate connections for residential subscribers over the local loop BUT UTP is susceptible to Interference so a solution is to use Cable TV.

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Introduction:

- Wireless communication is one of the fastest-growing technologies
- The demand for connecting devices without the use of cables is increasing everywhere
- Wireless LANs can be found on college campuses, in office buildings, and in many public areas

Architectural Comparison

- Architecture comparison of wired and wireless LANs

- ☐ Medium
- ☐ Hosts
- ☐ Isolated LANs
- ☐ Connection to other Networks
- ☐ Moving between Environments

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Characteristics of a Wireless LAN

- Several characteristics of wireless LANs either do not apply to wired LANs or the existence of these is negligible and can be ignored

- ☐ Attenuation
- ☐ Interference
- ☐ Multipath Propagation
- ☐ Error

Access Control:

- Most important issue in a wireless LAN is how a wireless host can get access to the shared medium (air)
- CSMA/CD does not work in wireless LANs for three reasons:
 1. Wireless hosts don't have power to send and receive at the same time
 2. The hidden station problem prevents collision detection
 3. The distance between stations can be large

IEEE 802.11 PROJECT :

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers
- It is sometimes called Wireless Ethernet
- The term WiFi (short for wireless fidelity) as a synonym for wireless LAN (certified by WiFi alliance)

Architecture:

- The standard defines two kinds of services:
 - ☐ The basic service set (BSS); and
 - ☐ The Extended service set (ESS)

Types of stations:

- No-Transition Mobility

- BSS-Transition Mobility
- ESS-Transition Mobility

MAC Sub layer:

- IEEE 802.11 defines two MAC sub-layers:
 - The Distributed Coordination Function (DCF) ; and
 - The Point Coordination Function (PCF)

Frame Types:

- Management Frames
- Control Frames
- Data Frames

Values of sub fields in control frame:

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Physical layers:

- All physical implementations, except the infrared, operate in the industrial, scientific, and medical (ISM) band, which defines 3 unlicensed bands in 3 ranges:
 - 902–928 MHz
 - 2.400–4.835 GHz
 - 5.725–5.850 GHz

BLUETOOTH:

- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other
- A Bluetooth LAN is an ad hoc network
- The devices, sometimes called gadgets, find each other and make a network called a Piconet

Architecture:

- Bluetooth defines two types of networks:
 - Piconet
 - Scatternet

Bluetooth Layers:

- Bluetooth uses several layers that do not exactly match those of the Internet model we have defined in this book

Bluetooth Devices:

- A Bluetooth device has a built-in short-range radio transmitter
- The current data rate is 1 Mbps with a 2.4-GHz bandwidth
- This means that there is a possibility of interference between the IEEE 802.11b wireless LANs and Bluetooth LANs

CH16

Connecting Devices:

- Hosts and networks do not normally operate in isolation
- Connecting devices connect hosts together to make a network or connect networks together to make an internet

- Connecting devices can operate in different layers of the Internet model
- Three kinds of connecting devices:

- ☐ Hubs
- ☐ Link-layer switches
- ☐ Routers

Hubs:

- Hub is a device that operates only in the physical layer
- Signals that carry information within a network can travel a fixed distance before attenuation impacts the data
- A hub (repeater) receives a signal and, before it becomes too weak or corrupted, regenerates it
- Hub is a device that operates only in the physical layer

Link-Layer Switches:

- A link-layer switch (or switch) operates in both the physical and the data-link layers
- As a physical-layer device, it regenerates the signal it receives
- As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame

Switch vs Hub:

- Switch has the 'Filtering' capability
- Unlike hub, a switch can check the destination address of a frame and decide on outgoing port

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- Switch eliminates collisions and does not require carrier sensing
- Switches connect heterogeneous devices

Link-layer Switches:

- A link-layer switch (or switch) operates in both the physical and the data-link layers

Loop Problem in Switch:

- Redundant switches create Loops in the system
- Created when two or more broadcasting LANs are connected by more than one switch

Spanning Tree Algorithm:

- In graph theory, Spanning Tree is a graph in which there is no loop
- In a switched LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop)
- To find the spanning tree, we assign a cost (metric) to each LAN link

Routers:

- We compare routers to two-layer switch and a hub
- A router is a three-layer device; it operates in the physical, data-link, and network layers

Router Vs Switch:

Three differences between a router and a repeater or a switch:

1. A router has a physical and logical (IP) address for each of its interfaces.
2. A router acts only on those packets in which the link-layer destination address matches the address of the interface at which the packet arrives.
3. A router changes the link-layer address of the packet (both source and destination) when it

forwards the packet.

Virtual LANS (VLAN):

- A VLAN is a LAN configured by software, not by physical wiring
- A station is considered part of a LAN if it physically belongs to that LAN i.e. The criterion of membership is geographic
- Provides a virtual connection between two stations belonging to two different physical LANs
- Membership of a VLAN:
- What characteristic can be used to group stations in a VLAN?
- Vendors use different characteristics such as interface numbers, port numbers, MAC addresses,

IP addresses, IP multicast addresses, or a combination of two or more of these

- Configuration of a VLAN:
- How are the stations grouped into different VLANs?
- Stations are configured in one of three ways:

- ☐ Manually
- ☐ Semi-Automatically
- ☐ Automatically

Communication between Switches:

- In a multi-switched backbone, each switch must know:
 - ☐ Which station belongs to which VLAN; and
 - ☐ The membership of stations connected to other switches

Advantages of Using VLANs:

- Cost and Time Reduction
- Creating virtual Workgroups
- Security

Comparison of Modern Access Technologies

- Telco

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- HFC
- FTTx

Fiber to Curb(FTC)

- An access network in which fiber is used for part, but not the entire link from the provider to the end-user
- An optical to electrical (O/E) conversion takes place somewhere near the end-user
- The terminal network segment of a FTTC network is usually twisted pair or coaxial cable
- The final optical receiver in a FTTC network typically serves several customers

Fiber To The Home (FTTH):

- Need: High-speed data, reliable voice and high-quality video
- Problems:
 - ☐ How to get high speed lines out to each customer?
 - ☐ How to future-proof the architecture?

Solution: FTTH

- Fiber-to-the-home (FTTH) is the installation of optical fiber from a telephone switch directly into the subscriber's home
- It is one of the latest access technologies
- FTTH is also referred to as Fiber-to-the-Building (FTTB)___

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Data Communication Topics

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Ch01. Characteristics of a Data Communication System

Ch01. Components of a Data Communication System

Ch01. Data Representation and Data Flow

Ch01. Networks

Ch01. Physical Structures

Ch01. Physical Topologies

Ch01. Local Area Network

Ch01. Wide Area Network

Ch01. Switching

Ch01. The Internet

Ch01. Internet History

Ch01. Internet Standards and Administration

Ch02. Protocol Layering - Introduction

Ch02. Protocol Layering - Advantages and Disadvantages

Ch02. Protocol Layering - Principles

Ch02. TCP/IP Protocol Suite

Ch02. TCP/IP Suite - Function of layers

Ch02. TCP/IP Suite - Layer Description

Ch02. Encapsulation and Decapsulation

Ch02. The Open System Interconnection (OSI) Model

Ch02. Data Communication vs Computer Networks

Ch03. Communication at Physical Layer

Ch03. Analog and Digital Signals

Ch03. Periodic and Non Periodic Analog Signals

Ch03. Sine Wave - Frequency

Ch03. Phase or Phase Shift
Ch03. Wavelength
Ch03. Time and Frequency Domains
Ch03. Composite Signals
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Ch03. Broadband Transmission (Modulation)
Ch03. Transmission Impairments
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Ch1 Introduction

- **Data communications** are the transfer of data from one device to another via some form of transmission medium.
- A data communications system must transmit data to the correct destination in an accurate and timely manner. The effectiveness of a data communications system depends on four fundamental characteristics: **delivery, accuracy, timeliness, and jitter**.
 - o **Jitter** refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.
- The **five components** that make up a data communications system are the message, sender, receiver, medium, and protocol.
- Text, numbers, images, audio, and video are different forms of information.
- Data flow between two devices can occur in one of three ways: **simplex, half-duplex, or full-duplex**.

o Simplex

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. Keyboards and traditional monitors are examples of simplex devices.

o Half-Duplex

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. Walkie-talkies and CB (citizens band) radios are both half-duplex systems

o **Full Duplex**

In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously. One common example of full-duplex communication is the telephone network

- **A network** is a set of communication devices connected by media links.

- **Network Criteria:** A network must be able to meet a certain number of criteria. The most important

of these are performance, reliability, and security.

- o **Performance** can be measured in many ways, including transit time and response time.

- o **Transit time** is the amount of time required for a message to travel from one device to another. **Response time** is the elapsed time between an inquiry and a response.

- o **Reliability:** In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure

- **PHYSICAL STRUCTURE (CONNECTIONS)**

- o **Types of connections:** In a point-to-point connection, two and only two devices are connected by a dedicated link. In a multipoint connection, three or more devices share a link.

- **Topology** refers to the physical or logical arrangement of a network. Devices may be arranged in a

mesh, star, bus, or ring topology.

1. **Mesh:** In a mesh topology, every device has a dedicated point-to-point link to every other device.

In mesh topology we need $n(n-1)/2$ duplex mode links.

2. **Star:** In a star topology, each device has a dedicated *point-to-point link only to a central controller*, usually called a hub.

3. **Bus:** A bus topology, on the other hand, is *multipoint*. One long cable acts as a *backbone* to link

all the devices in a network

4. **Ring:** In a ring topology, each device has a *dedicated point-to-point connection* with only the two

devices on either side of it.

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- **A network** can be categorized as a local area network, metropolitan area network or a wide area

network.

- o **A LAN** is a data communication system within a building, plant, or campus, or between

nearby buildings.

- o **A MAN** is a data communication system inside a city or town e.g cable network or DSL.
- o **A WAN** is a data communication system spanning states, countries, or the whole world.
- An internet is a network of networks.
- The Internet is a collection of many separate networks.
- There are local, regional, national, and international Internet service providers.
- **A protocol** is a set of rules that govern data communication; the key elements of a protocol are syntax, semantics, and timing.

The key elements of a protocol are **syntax, semantics, and timing**.

- o **Syntax**. The term syntax refers to the structure or format of the data, meaning the order in which they are presented.
- o **Semantics**. The word semantics refers to the meaning of each section of bits.
- How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation?
- **Standards** are necessary to ensure that products from different manufacturers can work together as expected.
- Data communication standards fall into two categories: **de facto** (meaning "by fact" or "by convention") and **de jure** (meaning "by law" or "by regulation")
- The ISO, ITD-T, ANSI, IEEE, and EIA are some of the organizations involved in standards creation.
- Forums are special-interest groups that quickly evaluate and standardize new technologies.
- A Request for Comment is an idea or concept that is a precursor to an Internet standard.

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Ch2 Network Modes

- The International Standards Organization (1947) created a model called the **Open Systems Interconnection OSI** (late 1970), which allows diverse systems to communicate.
 - The seven-layer OSI model provides guidelines for the development of universally compatible networking protocols.
 - The physical, data link, and network layers are the **network support layers**.
 - The session, presentation, and application layers are the **user support layers**.
 - The transport layer **links** the network support layers and the user support layers.
1. **The physical layer** coordinates the functions required to transmit a bit stream over a physical medium.
 2. **The data link layer** is responsible for delivering data units from one station to the next without errors.
- o **Data rate**: The transmission rate-the number of bits sent each second-is also defined by the physical layer.
 - o **Framing**: The data link layer divides the stream of bits received from the network layer into manageable data units called frames.

- o **Error control:** The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames.
- o At each layer, a **header**, or possibly a **trailer**, can be added to the data unit. Commonly, the **trailer** is added only at layer 2.

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3. **The network layer** is responsible for the source-to-destination delivery of a packet across multiple network links.

4. **The transport layer** is responsible for the process-to-process delivery of the entire message.

5. **The session layer** establishes, maintains, and synchronizes the interactions between communicating devices.

6. **The presentation layer** ensures interoperability between communicating devices through transformation of data into a mutually agreed upon format.

7. **The application layer** enables the users to access the network.

- **Encapsulation:** A packet (header and data) at level 7 is encapsulated in a packet at level 6. The whole packet at level 6 is encapsulated in a packet at level 5, and so on. In other words, the data portion of a packet at level $N - 1$ carries the whole packet (data and header and maybe trailer) from level N . The concept is called *encapsulation*; level $N - 1$ is not aware of which part of the encapsulated packet is data and which part is the header or trailer. For level $N - 1$, the whole packet coming from level N is treated as one integral unit.

- To carry sensitive information, a system must be able to ensure privacy. **Encryption** means that the sender transforms the original information to another form and sends the resulting message out over the network.

- **Decryption** reverses the original process to transform the message back to its original form.

- **TCP/IP is a five-layer** hierarchical protocol suite developed before the OSI model.

- The TCP/IP application layer is equivalent to the combined session, presentation, and application layers of the OSI model.

- Four levels of addresses are used in an internet following the TCP/IP protocols: physical (link) addresses, logical (IP) addresses, port addresses, and specific addresses.

- The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN.

- The IP address uniquely defines a host on the Internet.

- The port address identifies a process on a host.

- A specific address is a user-friendly address.

Physical Addresses

The physical address, also known as the link address, is the address of a node as defined by its LAN or

WAN. It is included in the frame used by the data link layer. It is the lowest-level address.

Logical addresses

A universal addressing system is needed in which each host can be identified uniquely, regardless of the underlying physical network. The logical addresses are designed for this purpose. A logical address in the

Internet is currently a 32-bit address that can uniquely define a host connected to the Internet.

Ch3 Data and Signals

- Data must be transformed to electromagnetic signals to be transmitted.
- **Data** can be analog or digital. **Analog data** are continuous and take continuous values. **Digital data**

have discrete states and take discrete values.

- **Signals** can be analog or digital. **Analog signals** can have an infinite number of values in a range;

digital signals can have only a limited number of values.

- In data communications, we commonly use **periodic analog signals** and **non periodic(aperiodic)**

digital signals

- **A periodic signal** completes a pattern within a measurable time frame, called a period, and repeats

that pattern over subsequent identical periods. **A non periodic signal** changes without exhibiting a

pattern or cycle that repeats over time.

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PERIODIC ANALOG SIGNALS

Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a sine

wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

- **The sine wave** is the most fundamental form of a periodic analog signal. A sine wave can be represented by three parameters: the peak amplitude, the frequency, and the phase.

o **Peak Amplitude:** The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries.

o **Frequency and period** are the *inverse* of each other. Frequency is the rate of change with respect to time. $f=1/T$

o **Phase** describes the position of the waveform relative to time.

Wavelength = propagation speed/ frequency =>

Time and Frequency Domains

- A complete sine wave in the time domain can be represented by *one single spike* in the frequency domain.

- A single-frequency sine wave is not useful in data communications; we need to send a composite

signal, a signal made of many simple sine waves.

- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

- **The bandwidth of a composite signal** is the difference between the highest and the lowest frequencies contained in that signal.

Digital Signal

- A digital signal is a composite analog signal with an infinite bandwidth.

- o **The bit rate** is the number of bits sent in 1s, expressed in bits per second (bps)

- o **Bit length** = propagation speed x bit duration

- Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only

if we have a low-pass channel with an infinite or very wide bandwidth.

- If the available channel is a bandpass channel, we cannot send a digital signal directly to the channel;

we need to convert the digital signal to an analog signal before transmission.

TRANSMISSION IMPAIRMENT

- *Attenuation, distortion, and noise* can impair a signal.

- **Attenuation** is the loss of a signal's energy due to the resistance of the medium.

- **Distortion** is the alteration of a signal due to the differing propagation speeds of each of the frequencies that make up a signal.

- **Noise** is the external energy that corrupts a signal.

- The bandwidth-delay product defines the number of bits that can fill the link.

DATA RATE LIMITS

Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

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PERFORMANCE

Network performance is measured by **bandwidth, throughput, latency and jitter**

Throughput

The throughput is a measure of how fast we can actually send data through a network.

Latency (Delay)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency = propagation time + transmission time + queuing time + processing delay

Jitter: Another performance issue that is related to delay is jitter. We can roughly say that jitter is a

problem if different packets of data encounter different delays and the application using the data at the

receiver site is time-sensitive (audio and video data, for example).

Bit rate for noiseless Channel: $\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$

Q: Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels.

The maximum bit rate can be calculated as

$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$

Ch4 Digital Transmission

- **Digital-to-digital conversion** involves three techniques: line coding, block coding, and scrambling.

- o **Line coding** is the process of converting digital data to a digital signal. We can roughly divide line coding schemes into five broad categories: unipolar, polar, bipolar, multilevel, and multi transition.

The data rate is sometimes called the **bit rate**; the signal rate is sometimes called the pulse rate, the modulation rate, or the **baud rate**.

Line Coding Schemes

- o **Block coding** provides redundancy to ensure synchronization and inherent error detection. Block coding is normally referred to as mB/nB coding; it replaces each m-bit group with an n-bit group.

- o **Scrambling** provides synchronization without increasing the number of bits. Two common scrambling techniques are *B8ZS* and *HDB3*.

_ Line Coding Schemes

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Unipolar Scheme

In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below.

- o **NRZ (Non-Return-to-Zero)** Traditionally, a unipolar scheme was designed as a non-return-to-zero

(NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0.

It is called NRZ because the signal does not return to zero at the middle of the bit

Polar Schemes

In polar schemes, the voltages are on the both sides of the time axis. For example, the voltage level for 0

can be positive and the voltage level for 1 can be negative

- o **Non-Return-to-Zero (NRZ)** In polar NRZ encoding, we use two levels of voltage amplitude.

We can have two versions of polar NRZ: NRZ-L and NRZ-I. The figure also shows the value of r ,

the average baud rate, and the bandwidth. In the first variation, NRZ-L (NRZ-Level), the level of

the voltage determines the value of the bit. In the second variation, NRZ-I (NRZ-Invert), the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1

- o **Return to Zero (RZ)** The main problem with NRZ encoding occurs when the sender and

receiver clocks are not synchronized. The receiver does not know when one bit has ended and the next bit is starting. One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero. In RZ, the signal changes not between bits but during the bit.

o **Biphase:** Manchester and Differential Manchester The idea of RZ (transition at the middle of the bit) and the idea of NRZ-L are combined into the Manchester scheme. In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization. Differential Manchester, on the other hand, combines the ideas of RZ and NRZ-I.

Bipolar Schemes

In bipolar encoding (sometimes called multilevel binary), there are three voltage levels: positive, negative, and zero. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.

Multilevel Schemes

The desire to increase the data speed or decrease the required bandwidth has resulted in the creation of many schemes. The goal is to increase the number of bits per baud by encoding a pattern of m data elements into a pattern of n signal elements. We only have two types of data elements (0s and 1s), which means that a group of m data elements can produce a combination of 2^m data patterns.

Multiline Transmission: MLT-3

The multiline transmission, three level (MLT-3) scheme uses three levels (+V, 0, and -V) and three transition rules to move between the levels.

1. If the next bit is 0, there is no transition.
2. If the next bit is 1 and the current level is not 0, the next level is 0.
3. If the next bit is 1 and the current level is 0, the next level is the opposite of the last nonzero level.

Scrambling

R8ZS

Bipolar with S-zero substitution (BSZS) is commonly used in North America. In this technique, eight consecutive zero-level voltages are replaced by the sequence OOOVBOVB.

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HDB3

High-density bipolar 3-zero (HDB3) is commonly used outside of North America. In this technique, which is more conservative than B8ZS, four consecutive zero-level voltages are replaced with a sequence of OOOV or BOO

• Analog to Digital Conversion

1. **PCM:** The most common technique to change an analog signal to digital data (digitization) is called **pulse code modulation (PCM)**.

Components of PCM encoder

- i. The analog signal is **sampled**. (Sampling)
- ii. The sampled signal is **quantized**. (Quantizing)
- iii. The quantized values are **encoded** as streams of bits. (Encoding)

- The first step in PCM is sampling. The analog signal is sampled every T_s , where T_s is the sample

interval or period. The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by f_s . There are three **sampling methods**—*ideal*, *natural*, and *flat-top*.

- According to the **Nyquist theorem**, to reproduce the original analog signal, one necessary condition

is that the sampling rate be at least twice the highest frequency in the original signal.

- Other sampling techniques have been developed to reduce the complexity of PCM.

- The simplest is **delta modulation**. PCM finds the value of the signal amplitude for each sample; DM

finds the change from the previous sample.

Bit rate = sampling rate x number of bits per sample

2. Delta Modulation (DM)

PCM is a very complex technique. Other techniques have been developed to reduce the complexity of PCM. The simplest is **delta modulation**. PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample.

Components of Delta Modulation

- i. Comparator
- ii. The Delay Unit
- iii. Staircase Maker

DATA TRANSMISSION MODES

- While there is only one way to send **parallel data**, there are three subclasses of **serial transmission**:

asynchronous, synchronous, and isochronous.

- o In **asynchronous transmission**, we send 1 start bit (0) at the beginning and 1 or more stop bits (1 s) at the end of each byte.

- o In **synchronous transmission**, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

- o The **isochronous mode** provides synchronized for the entire stream of bits must. In other words, it guarantees that the data arrive at a fixed rate.

Ch5 Analog Transmission

_ **Digital-to-analog conversion** is the process of changing one of the characteristics of an analog signal

based on the information in the digital data.

- Digital-to-analog conversion can be accomplished in several ways: **amplitude shift keying (ASK)**,

frequency shift keying (FSK), and **phase shift keying (PSK)**.

- **Quadrature amplitude modulation (QAM)** combines ASK and PSK.

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- In **amplitude shift keying**, the amplitude of the carrier signal is varied to create signal elements. Both

frequency and phase remain constant while the amplitude changes.

- In **frequency shift keying**, the frequency of the carrier signal is varied to represent data. The frequency of the modulated signal is constant for the duration of one signal element, but changes for

the next signal element if the data element changes.

Both peak amplitude and phase remain constant for all signal elements.

- In **phase shift keying**, the phase of the carrier is varied to represent two or more different signal

elements. Both peak amplitude and frequency remain constant as the phase changes.

- A **constellation diagram** shows us the amplitude and phase of a signal element, particularly when we

are using two carriers (one in-phase and one quadrature).

- **Quadrature amplitude modulation (QAM)** is a combination of ASK and PSK.

QAM uses two carriers, one in-phase and the other quadrature, with different amplitude levels for

each carrier.

_ **Analog-to-analog conversion** is the representation of analog information by an analog signal. Conversion is needed if the medium is band pass in nature or if only a band pass bandwidth is available to us.

Analog-to-analog conversion can be accomplished in three ways: **amplitude modulation (AM)**,

frequency modulation (FM), and **phase modulation (PM)**.

- o In **AM transmission**, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same; only the amplitude changes to follow variations in the information.

- o In **FM transmission**, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.

- o In **PM transmission**, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the

phase of the carrier changes correspondingly.

Ch6 Bandwidth Utilization (Multiplexing and Spreading)

- **Bandwidth utilization** is the use of available bandwidth to achieve specific goals. In multiplexing, our goal is efficiency; we combine several channels into one. In spreading, our goals are privacy and

anti-jamming

- Efficiency can be achieved by using multiplexing; privacy and anti jamming can be achieved by using spreading.

- **Multiplexing** is the set of techniques that allows the simultaneous transmission of multiple signals

across a single data link. In a multiplexed system, n lines share the bandwidth of one link. The word

link refers to the physical path. The word channel refers to the portion of a link that carries a transmission.

There are three basic multiplexing techniques: **frequency-division multiplexing, wavelengthdivision**

multiplexing, and **time-division multiplexing**. The first two are techniques designed for analog signals, the third, for digital signals

- o **Frequency-division multiplexing (FDM)** is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

- o **Wavelength-division multiplexing (WDM)** is designed to use the high bandwidth capability of fiber-optic cable. WDM is an analog multiplexing technique to combine optical signals.

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- o **Time-division multiplexing (TDM)** is a digital process that allows several connections to share the high bandwidth of a link. TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

Interleaving: TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the de-multiplexing side. This process is called interleaving.

- We can divide TDM into two different schemes: synchronous or statistical. In **synchronous TDM**, each input connection has an allotment in the output even if it is not sending data. In **statistical TDM**, slots are dynamically allocated to improve bandwidth efficiency.

- **Data Rate Management:** One problem with TDM is how to handle a disparity in the input data rates.

In all our discussion so far, we assumed that the data rates of all input lines were the same. However,

if data rates are not the same, three strategies, or a combination of them, can be used.

We call these three strategies **multilevel multiplexing**, **multiple-slot allocation**, and **pulse stuffing**.

- In **spread spectrum (SS)**, we combine signals from different sources to fit into a larger bandwidth.

Spread spectrum is designed to be used in wireless applications in which stations must be able to

share the medium without interception by an eavesdropper and without being subject to jamming

from a malicious intruder.

- The **frequency hopping spread spectrum (FHSS)** technique uses M different carrier frequencies

that are modulated by the source signal. At one moment, the signal modulates one carrier frequency;

at the next moment, the signal modulates another carrier frequency.

- The **direct sequence spread spectrum (DSSS)** technique expands the bandwidth of a signal by

replacing each data bit with n bits using a spreading code. In other words, each bit is assigned a code

of n bits, called chip.

Ch7 Transmission Media

- **Transmission media** lie below the physical layer.

1. A **guided medium** provides a physical conduit from one device to another. Twisted pair cable,

coaxial cable, and optical fiber are the most popular types of guided media.

- **Twisted-pair cable** consists of two insulated copper wires twisted together. Twisted pair cable is

used for voice and data communications.

- **Coaxial cable** consists of a central conductor and a shield. Coaxial cable can carry signals of higher

frequency ranges than twisted-pair cable. Coaxial cable is used in cable TV networks and traditional

Ethernet LANs.

- **Fiber-optic cables** are composed of a glass or plastic inner core surrounded by cladding, all enclosed

in an outside jacket. Fiber-optic cables carry data signals in the form of light. The signal is propagated

along the inner core by reflection. Fiber optic transmission is becoming increasingly popular due to

its noise resistance, low attenuation, and high-bandwidth capabilities. Fiber-optic cable is used in

backbone networks, cable TV networks, and Fast Ethernet networks.

Propagation Modes

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_ **Multimode** is so named because multiple beams from a light source move through the core in different paths.

_ In **multimode step-index fiber**, the density of the core remains constant from the center to the edges.

A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.

_ A second type of fiber, called **multimode graded-index fiber**, decreases this distortion of the signal through the cable.

_ **Single-Mode:** Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.

2. Unguided media (free space) transport electromagnetic waves without the use of a physical conductor.

• **Unguided (Wireless) data** are transmitted through ground propagation, sky propagation, and line-of-sight

propagation. This type of communication is often referred to as wireless communication.

o **In ground propagation**, radio waves travel through the lowest portion of the atmosphere, hugging the earth.

o **In sky propagation**, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth.

o **Line-of-sight propagation** is tricky because radio transmissions cannot be completely focused.

_ **Wireless waves** can be classified as **radio waves, microwaves, or infrared waves**.

Radio waves are *omni-directional*; microwaves are *unidirectional*.

_ **Microwaves** are used for cellular phone, satellite, and wireless LAN communications.

_ **Infrared waves** are used for short-range communications such as those between a PC and a peripheral device. It can also be used for indoor LANs.

Ch8 Switching

• **A switched network** consists of a series of interlinked nodes, called switches. Traditionally, three

methods of switching have been important: circuit switching, packet switching, and message switching.

o We can divide today's networks into three broad categories: circuit-switched networks, packet-switched networks, and message-switched.

o Packet-switched networks can also be divided into two subcategories: virtual-circuit networks and datagram networks

• **A circuit-switched network** is made of a set of switches connected by physical links, in which each

link is divided into n channels. Circuit switching takes place at the *physical layer*. In circuit

switching, the resources need to be reserved during the setup phase; the resources remain dedicated

for the entire duration of data transfer phase until the teardown phase.

Three Phases

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The actual communication in a circuit-switched network requires three phases: **connection setup**,

data transfer, and **connection teardown**.

- o **Connection Setup Phase:** Connection setup means creating dedicated channels between the switches.

- o **Data Transfer Phase:** After the establishment of the dedicated circuit (channels), the two parties can transfer data.

- o **Teardown Phase:** When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

- **In packet switching**, there is no resource allocation for a packet. This means that there is no reserved

bandwidth on the links, and there is no scheduled processing time for each packet. Resources are

allocated on demand.

- o **In a datagram network**, each packet is treated independently of all others. Packets in this approach are referred to as datagrams. There are no setup or teardown phases.

- o **A virtual-circuit network** is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

- **In message switching**, each switch stores the whole message and forwards it to the next switch. We

will not discuss this topic in this book.

Structure of Circuit Switches

Circuit switching uses either of two technologies: the **space-division switch** or the **time-division switch**.

switch.

o Space-Division Switch

In space-division switching, the paths in the circuit are separated from one another spatially.

o Time-Division Switch

Time-division switching uses time-division multiplexing (TDM) inside a switch. The most popular

technology is called the time-slot interchange (TSI).

- A switch in a packet-switched network has a different structure from a switch used in a circuit switched network. We can say that a packet switch has four types of components: input ports, output

ports, a routing processor, and switching fabric.

SOLVED CURRENT FINAL TERM PAPERS

CS601 DATA COMMUNICATION

BY SONU MUGHAL

27/2/19

Lan protocols

There are many **LAN protocols** in use today. Some of the more common ones are Ethernet, Asynchronous Transfer Mode (ATM), Token Ring and Fiber Distributed Data Interface (FDDI). Ethernet is a common **LAN protocol** that can be found in most organizations. ATM is gaining popularity.

What are the LAN WAN protocols?

Wan/Lan Protocols. ... The difference between a **LAN** and a **WAN** is the distance the data travels, with a **LAN** typically serving a single building and a **WAN** usually covering different geo graph locations. Two systems must be using the same **LAN** or **WAN protocol** to communicate.

What is meant by network protocol?

Network protocols are formal standards and policies comprised of rules, procedures and formats that **define** communication between two or more devices over a **network**. **Network protocols** govern the end-to-end processes of timely, secure and managed data or **network** communication.

What are the different types of protocols?

Types of Protocols

- • TCP. Transmission control protocol is used for communication over a network. ...
- • Internet Protocol (IP) IP is also working with TCP. ...
- • FTP. File transfer protocol is basically used for transferring files to different networks.
- ...
- • SMTP. ...
- • HTTP. ...
- • Ethernet. ...
- • Telnet. ...
- • Gopher.

What network protocol is commonly used on LANs?

Ethernet is by far the most commonly used LAN technology. **Token Ring** technology is still used by some companies. **FDDI** is sometimes used as a backbone LAN interconnecting **Ethernet** or **Token Ring** LANs.

How checksum work

1. A **checksum** is a small-sized datum derived from a block of digital data for the purpose of detecting errors that may have been introduced during its transmission or storage. ...
2. The actual procedure which yields the **checksum** from a data input is called a **checksum function** or **checksum**

A **checksum** is a small-sized datum derived from a block of digital data for the purpose of detecting errors that may have been introduced during its transmission or storage. It is usually applied to an installation file after it is received from the download server. By themselves, checksums are often used to verify data integrity but are not relied upon to verify data authenticity.

The actual procedure which yields the checksum from a data input is called a **checksum function** or **checksum algorithm**. Depending on its design goals, a good checksum algorithm will usually output a significantly different value, even for small changes made to the input. This is especially true of cryptographic hash functions, which may be used to detect many data corruption errors and verify overall data integrity; if the computed checksum for the current data input matches the stored value of a previously computed checksum, there is a very high probability the data has not been accidentally altered or corrupted.

Inverse TDM

Inverse **multiplexing** speeds up data transmission by dividing a data stream into multiple **concurrent** streams that are **transmitted** at the same time across separate channels (such as a T-1 or E-1 lines) and are then **reconstructed** at the other end back into the original data stream

. **What are network access methods?**

Indexed **access methods** are widely used to keep track of records in a file and files on disk. ... See ISAM, direct **access method** and sequential **access method**. Communications **Access Methods**. Local area **network (LAN) access methods**, such as CSMA/CD in Ethernet, transfer data to and from connected computers on the **network**.

three control access method names

In controlled access, the stations seek information from one another to find which station has the right to send. It allows only one node to send at a time, to avoid collision of messages on shared medium. The three controlled-access methods are:

1. Reservation
2. Polling
3. Token Passing

Bandwidth of frequency modulation

The **bandwidth** of an **FM** signal is not as straightforward to calculate as that of an AM signal. To take the example of a typical broadcast **FM** signal that has a deviation of $\pm 75\text{kHz}$ and a maximum **modulation frequency** of 15 kHz, the **bandwidth** of 98% of the power approximates to $2(75 + 15) = 180\text{kHz}$.

What is the bandwidth of AM signal?

This is the range of carrier frequencies available. The information transmitted is music and talk which falls in the audio spectrum. The full audio spectrum ranges up to **20 kHz**, but AM radio limits the upper modulating frequency to **5 kHz**. This results in a maximum bandwidth of **10 kHz**.

How is bandwidth related to frequency?

The higher the **frequency**, the more **bandwidth** is available. For example, at 100KHz (**frequency**), a signal can run from 0 to 200KHz. ... But there's no

direct **relationship** between **frequency** and **bandwidth**. A 20MHz channel is 20 MHz wide (its **bandwidth**), whether it's at 2.4GHz or 5GHz (its **frequency**)

Why FM is called constant bandwidth system?

The simplest answer is that "Because the **bandwidth** does not increase quickly as compared to the modulation frequency in the **FM**". **FM is called a constant bandwidth system** because: ... The amplitude of a frequency modulated signal remains **constant**.

What is bandwidth modulation?

The **modulation bandwidth** is defined as the maximum rate of change in output frequency that can be attained using control voltage a vco **bandwidth**, spectrum and sidebands are great importance when.

What is the bandwidth of FM?

The bandwidth of an FM signal is not as straightforward to calculate as that of an AM signal. To take the example of a typical broadcast FM signal that has a deviation of $\pm 75\text{kHz}$ and a maximum modulation frequency of **15 kHz**, the bandwidth of 98% of the power approximates to $2(75 + 15) = \mathbf{180\text{kHz}}$.

What is the difference between frequency and bandwidth?

Bandwidth and **frequency** both are the measuring terms of networking. The basic **difference between bandwidth** and **frequency** is that **bandwidth** measures the amount of data transferred per second whereas the **frequency** measure the number of oscillations of the data signal per second.

Impacted bits calculate by burst error

The term **burst error** means that two or more **bits** in the data unit have **changed** from 0 to 1 or vice-versa. Note that **burst error** doesn't necessary means that **error**

occurs in consecutive **bits**. The length of the **burst error** is measured from the first corrupted **bit** to the last corrupted **bit**.

What is burst error with example?

In telecommunication, a **burst error** or **error burst** is a contiguous sequence of symbols, received over a communication channel, such that the first and last symbols are in **error** and there exists no contiguous subsequence of m correctly received symbols within the **error burst**.

What is the difference between single bit error and burst error?

Single-bit errors are least likely type of **error** because their duration or noise is normally longer than duration of 1 **bit**. When more than a **single bit** of data unit gets corrupted it is known as **Burst error**. In **comparison** of **single-bit errors**, **burst errors** are more likely to occur.

What is burst error in data communication?

In telecommunication, a **burst error** or **error burst** is a contiguous sequence of symbols, received over a **communication** channel, such that the first and last symbols are in **error** and there exists no contiguous subsequence of m correctly received symbols within the **error burst**.

what is LAN

LAN is a short form of local area network. Local area network is a network for connecting computers and other computer accessories with each other. Computer accessories include printers, scanners, game consoles etc. LAN is used to make the connection of computers within one building. To make the connection between the computers we use communication devices and cables. Communication devices are hubs, switches, and routers. Ethernet cables are connected to hubs, switches, and routers by rj45 ports.

What is a virtual local area network (VLAN)?

In a virtual area network, different switches of LAN are logically connected to each other to make the single switch. This type of network is used to logically differentiate and limit the different level of computers. For example, there are different departments in the office like the accounting department, support

department, and administrative department. All the computers of these departments are logically connected to the same switch, but one department computer cannot access another department computer. So same department computers can communicate with each other, but different department computers cannot communicate with each other.

Advantage of LAN

In LAN computers can exchange data and messages in the easy and **fast** way. It also saves time and makes our work **fast**. Every user can share messages and data with any other user on LAN. The user can log in from any computer on the network and access the same data placed on the server.

.Names of data link protocol

data link protocol - Computer Definition. In **networking** and communications, the transmission of a unit of **data** (frame, packet) from one node to another. Known as a "**layer 2 protocol**," the **data link protocol** is responsible for ensuring that the bits and bytes received are identical to the bits and bytes sent

What are the protocols of data link layer?

The data link layer is the protocol layer in a program that handles the moving of data into and out of a physical link in a network. The data link layer is Layer 2 in the **Open Systems Interconnection (OSI)** architecture model for a set of telecommunication protocols.

What are the 7 network layers?

The International Standards Organization (ISO) developed the Open Systems Interconnection (OSI) model. It divides **network** communication into **seven layers**. **Layers** 1-4 are considered the lower **layers**, and mostly concern themselves with moving data around. **Layers** 5-7, the upper **layers**, contain application-level data

How many layers TCP IP protocol has?

In some sites , it is given that top/it has **4 layers**, network access,internet,transport,application. But in some other references , it is given that it has 5 layers (instead of network access, physical and link.

What are the 4 layers of TCP/IP?

The **OSI** reference model has seven layers: **physical**, **data link**, **network**, **transport**, session, presentation, and application. The TCP/IP reference model has four layers: **Network** interface, Internet, **Transport**, and Application.

The 5-Layer Model (the Name TCP Model) Layer

4	Transport
3	Internet (sometimes called the Network Layer)
2	Network (sometimes called the Data Link Layer)
1	Physical

DATA COMMUNICATION
CS601

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Data?

The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.

What is communication?

When we communicate, we are sharing information. This sharing can be local or remote.

What is data communication?

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs)

What is telecommunication?

The term **telecommunication**, which includes telephony, telegraphy, and television, means communication at a distance (*tele* is Greek for “far”)

Characteristics of data communication system?

The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

1. Delivery. The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

2. Accuracy. The system must deliver the data accurately. Data that have been 2

altered in transmission and left uncorrected are unusable.

3. Timeliness. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called *real-time* transmission.

4. Jitter. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

Components of data communication system?

A data communications system has five components

1. Message. The **message** is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. Sender. The **sender** is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver. The **receiver** is the device that receives the message. It can be a computer, 3

workstation, telephone handset, television, and so on.

4. Transmission medium. The **transmission medium** is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation and data flow

Information today comes in different forms such as text, numbers, images, audio, and video.

Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex 4

What is network?

A **network** is the interconnection of a set of devices capable of communication

Physical Structures

Before discussing networks, we need to define some network attributes.

Type of Connection

A network is two or more devices connected through links.

Links: A link is a communication pathway that transfers data from one device to another.

There are two possible types of connections: **point-to-point** and **multipoint**.

Point-to-Point

A **point-to-point connection** provides a dedicated link between two devices

Multipoint

A **multipoint** (also called **multidrop**) **connection** is one in which more than two specific devices share a single link

Physical Topology

The term ***physical topology*** refers to the way in which a network is laid out physical the term ***physical topology*** refers to the way in which a network is laid out physical

Topology: Two or more devices connect to a link; two or more links form a topology

The topology

of a network is the geometric representation of the relationship of all the links and linking devices (usually called ***nodes***) to one another

There are four basic topologies

possible: mesh, star, bus, and ring.

Mesh Topology

In a **mesh topology**, every device has a dedicated point-to-point link to every other device.

The main disadvantages of a mesh are related to the amount of cabling and the 6

number of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive. For these reasons a mesh topology is usually implemented in a limited fashion

Star Topology

In a **star topology**, each device has a dedicated point-to-point link only to a central controller, usually called a **hub**. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices. A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. Other advantages include robustness. If one link fails, only that link is affected. All 7

other links remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

The star topology is used in local-area networks (LANs), as we will see in Chapter 13. High-speed LANs often use a star topology with a central hub.

Bus Topology

The preceding examples all describe point-to-point connections. A **bus topology**, on the other hand, is multipoint. One long cable act as a **backbone** to link all the devices in a network 8

Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.

Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies. In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated. Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable. Adding new devices may therefore require modification or replacement of the backbone.

In addition, a fault or break in the bus cable stops all transmission, even between 9

devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Ring Topology

In a **ring topology**, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along

A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified. Generally, in a ring a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location. 10

However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Ring topology was prevalent when IBM introduced its local-area network, Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

Classification of networks:

- • Size
- • Geographical
- • Coverage
- • ownership

Types of networks

There are two types of networks, **LANs** and **WANs**, we define switching, which is used to connect networks to form an **internetwork** (a network of networks).

Local area network (LAN)

- Usually Privately owned
- Connects some hosts in a single office, building, or campus
- **Can be as simple as two PCs and a printer in someone's home office**
- Can extend throughout a company
- Host Address

Wide area network WAN:

- Wider geographical span than a LAN
- Spans a town, a state, a country, or even the world
- **Interconnects connecting devices such as switches, routers, or modems**
- Normally created and run by communication companies
- **Point-to-Point WAN**
- Switched WAN
- Internetwork

Point-to-Point WAN

A point-to-point WAN is a network that connects two communicating devices through a transmission

media (cable or air). We will see examples of these WANs when we discuss how to connect the networks to one another. 12

- Switching

Switched WAN

A switched WAN is a network with more than two ends. A switched WAN, as we will see shortly, is used in the backbone of global communication today. We can say that a switched WAN is a combination of several point-to-point WANs that are connected by

- switches
- **Circuit-Switched Network**
- **Packet-Switched Network**

Packets? In a computer network, the communication between the two ends is done in blocks of data called **packets**.

Backbones and provider networks are also called **Internet Service Providers (ISPs)**. The backbones are often referred to as *international ISPs*; the provider networks are often referred to as *national* or *regional ISPs*.

Accessing the Internet

The Internet today is an internetwork that allows any user to become part of it. The user, however, needs to be physically connected to an ISP. The physical connection is normally done through a point-to-point WAN.

- **An internet (note the lowercase I) is two or more networks that can communicate with each other**

- **The Internet (uppercase I), and is composed of thousands of interconnected networks.**
- Accessing the Internet

Internet history:

- **Telegraph and Telephone networks, before 1960: ✓ Constant-rate communication only**
- ✓ **ARPANET- Packet Switched**

- Birth of the Internet & TCP/IP
- MILNET
- CSNET
- NSFNET
- Internet Today

Internet standard and administration:

- Internet draft
- Request for Comments (RFC) ✓ Proposed Standard

✓ Draft Standard

✓ Internet Standard

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✓ Historic

✓ Experimental

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✓ Informational

INTERNET HISTORY

Now that we have given an overview of the Internet, let us give a brief history of the Internet. This brief history makes it clear how the Internet has evolved from a private network to a global one in less than 40 years.

Early History

There were some communication networks, such as telegraph and telephone networks, before 1960. These networks were suitable for constant-rate communication at that time, which means that after a connection was made between two users, the encoded message 14

(telegraphy) or voice (telephony) could be exchanged. A computer network, on the other hand, should be able to handle *burst* data, which means data received at variable rates at different times. The world needed to wait for the packet-switched network to be invented.

Birth of Packet-Switched Networks

The theory of packet switching for bursty traffic was first presented by Leonard Kleinrock in 1961 at MIT. At the same time, two other researchers, Paul Baran at Rand Institute and Donald Davies at National Physical Laboratory in England, published some papers about packet-switched networks

ARPANET

In the mid-1960s, mainframe computers in research organizations were stand-alone devices. Computers from different manufacturers were unable to communicate with one another. The **Advanced Research Projects Agency (ARPA)** in the Department of Defense (DOD) was interested in finding a way to connect computers so that the researchers they funded could share their findings, thereby reducing costs and eliminating duplication of effort.

In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for the **Advanced Research Projects Agency Network (ARPANET)**, a small network of connected computers. The idea was that each host computer (not necessarily from the same manufacturer) would be attached to a specialized computer, called an *interface message processor* (IMP). The IMPs, in turn, would be connected to each other. Each IMP had to be able to communicate with other IMPs as well as with its 15

own attached host.

By 1969, ARPANET was a reality. Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of Utah, were connected via the IMPs to form a network. Software called the *Network Control Protocol* (NCP) provided communication between the hosts.

Birth of the Internet

In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the *Internetworking Project*. They wanted to link dissimilar networks so that a host on one network could communicate with a host on another. There were many problems to overcome: diverse packet sizes, diverse interfaces, and diverse transmission rates, as well as differing reliability requirements. Cerf and Kahn devised the idea of a device called a *gateway* to serve as the intermediary hardware to transfer data from one network to another.

TCP/IP

Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of data. This was a new version of NCP. This paper on transmission control protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway. A radical idea was the transfer of responsibility for error correction from the IMP to the host machine. This ARPA Internet now became the focus of the communication effort. Around this time, responsibility for the ARPANET was handed 16

over to the Defense Communication Agency (DCA).

In October 1977, an internet consisting of three different networks (ARPANET, packet radio, and packet satellite) was successfully demonstrated. Communication between networks was now possible.

Shortly thereafter, authorities made a decision to split TCP into two protocols: **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**. IP would handle datagram routing while TCP would be responsible for higher level functions such as segmentation, reassembly, and error detection. The new combination became known as TCP/IP.

In 1981, under a Defence Department contract, UC Berkeley modified the UNIX operating system to include TCP/IP. This inclusion of network software along with a popular operating system did much for the popularity of internetworking. The open (non-manufacturer-specific) implementation of the Berkeley UNIX gave every manufacturer a working code base on which they could build their products.

In 1983, authorities abolished the original ARPANET protocols, and TCP/IP became the official protocol for the ARPANET. Those who wanted to use the Internet to access a computer on a different network had to be running TCP/IP.

MILNET

In 1983, ARPANET split into two networks: **Military Network (MILNET)** for military users and ARPANET for nonmilitary users.

CSNET

Another milestone in Internet history was the creation of CSNET in 1981. **Computer** 17

Science Network (CSNET) was a network sponsored by the National Science Foundation (NSF). The network was conceived by universities that were ineligible to join ARPANET due to an absence of ties to the Department of Defense. CSNET was a less expensive network; there were no redundant links and the transmission rate was slower. By the mid-1980s, most U.S. universities with computer science departments were part of CSNET. Other institutions and companies were also forming their own networks and using TCP/IP to interconnect. The term *Internet*, originally associated with government-funded connected networks, now referred to the connected networks using TCP/IP protocols.

NSFNET

With the success of CSNET, the NSF in 1986 sponsored the **National Science Foundation Network (NSFNET)**, a backbone that connected five supercomputer centers located throughout the United States. Community networks were allowed access to this backbone, a T-1 line (see Chapter 6) with a 1.544-Mbps data rate, thus providing connectivity throughout the United States. In 1990, ARPANET was officially retired and replaced by NSFNET. In 1995, NSFNET reverted back to its original concept of a research network.

ANSNET

In 1991, the U.S. government decided that NSFNET was not capable of supporting the rapidly increasing Internet traffic. Three companies, IBM, Merit, and Verizon, filled the void by forming a nonprofit organization called Advanced Network & Services (ANS) 18

to build a new, high-speed Internet backbone called **Advanced Network Services Network (ANSNET)**.

Internet Today

Today, we witness a rapid growth both in the infrastructure and new applications. The Internet today is a set of peer networks that provide services to the whole world. What has made the Internet so popular is the invention of new applications

World Wide Web

The 1990s saw the explosion of Internet applications due to the emergence of the World Wide Web (WWW). The Web was invented at CERN by Tim Berners-Lee. This invention has added the commercial applications to the Internet.

Multimedia

Recent developments in the multimedia applications such as voice over IP (telephony), video over IP (Skype), view sharing (YouTube), and television over IP (PPLive) has increased the number of users and the amount of time each user spends on the network

STANDARDS AND ADMINISTRATION

In the discussion of the Internet and its protocol, we often see a reference to a standard or an administration entity. In this section, we introduce these standards and administration entities for those readers that are not familiar with them; the section can be skipped if the reader is familiar with them.

1.5.1 Internet Standards

An **Internet standard** is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet. It is a formalized regulation that must be followed. 19

There is a strict procedure by which a specification attains Internet standard status. A specification

begins as an Internet draft. An **Internet draft** is a working document (a work in progress) with no official status and a six-month lifetime. Upon recommendation from the Internet authorities, a draft may be published as a **Request for Comment (RFC)**. Each RFC is edited, assigned a number, and made available to all interested parties. RFCs go through maturity levels and are categorized according to their requirement level.

Internet Administration

The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity. Various groups that coordinate Internet issues have guided this growth and development. Appendix G gives the addresses, e-mail addresses, and telephone numbers for some of these groups.

Summary

Data communications are the transfer of data from one device to another via some form of transmission medium. A data communications system must transmit data to the correct destination in an accurate and timely manner. The five components that make up a data communications system is the message, sender, receiver, medium, and protocol. Text, numbers, images, audio, and video are different forms of information. Data flow between two devices can occur in one of three ways: simplex, half-duplex, or full-duplex.

A network is a set of communication devices connected by media links. In a point-to-point connection, two and only two devices are connected by a dedicated link. In a 20

multipoint connection, three or more devices share a link. Topology refers to the physical or logical arrangement of a network. Devices may be arranged in a mesh, star, bus, or ring topology.

A network can be categorized as a local area network or a wide area network. A LAN is a data communication system within a building, plant, or campus, or between nearby buildings. A WAN is a data communication system spanning states, countries, or the whole world. An internet is a network of networks. The Internet is a collection of many separate networks.

The Internet history started with the theory of packet switching for bursty traffic. The history continued when The ARPA was interested in finding a way to connect computers so that the researchers they funded could share their findings, resulting in the creation of ARPANET. The Internet was born when Cerf and Kahn devised the idea of a device called a *gateway* to serve as the intermediary hardware to transfer data from one network to another. The TCP/IP protocol suite paved the way for creation of today's Internet. The invention of WWW, the use of multimedia, and peer-to-peer communication helps the growth of the Internet.

An Internet standard is a thoroughly tested specification. An Internet draft is a working document with no official status and a six-month lifetime. A draft may be published as a Request for Comment (RFC). RFCs go through maturity levels and are categorized according to their requirement level. The Internet administration has evolved with the Internet. ISOC promotes research and activities. IAB is the technical

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advisor to the ISOC. IETF is a forum of working groups responsible for operational problems. IRTF is a forum of working groups focusing on long-term research topics.

Chp#02

Protocol layering

Introduction

What is protocol?

PROTOCOL LAYERING

We defined the term *protocol* in Chapter 1. In data communication and networking, a protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively. When communication is simple, we may need only one simple protocol; when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or **protocol layering**.

ADVANTAGE AND DISADVANTAGE

- Advantages ✓ Modularity

✓ Separation of Service & Implementation

✓ Reduced Complexity &

- Disadvantages

- None Really!

Cost 22

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Scenarios

Let us develop two simple scenarios to better understand the need for protocol layering.

Principles of Protocol Layering

Let us discuss two principles of protocol layering.

First Principle

The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction. For example, the third layer task is to listen (in one direction) and *talk* (in the other direction). The second layer needs to be able to encrypt and decrypt. The first layer needs to send and receive mail.

Second Principle

The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical. For example, the object under layer 3 at both sites should be a plaintext letter. The object under layer 2 at both sites should be a ciphertext letter. The object under layer 1 at both sites should be a piece of mail.

Logical Connections

logical (imaginary) connection

at each layer through which they can send the object created from that layer. 23

TCP/IP PROTOCOL SUITE

TCP/IP (Transmission

Control Protocol/Internet Protocol). TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. The term *hierarchical* means that each upper level protocol is supported by the services provided by one or more lower level protocols. The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a five-layer model.

- TCP/IP Protocol Suite ✓ Protocol suite used in Internet today
- ✓ Each Layer provides specific functionality
- ✓ Hierarchical Protocol
- ✓ Presented in 1973 and chosen to be the official protocol of Internet in 1983

Layered Architecture

To show how the layers in the TCP/IP protocol suite are involved in communication between two hosts, we assume that we want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch. We also assume that the links are connected by one router, 25

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Addressing in TCP/IP Protocol Suite

- Every communication needs at least two addresses: ✓ Source Address & ✓ Destination Address
- ✓ Addressing by Layer
- ✓ **Physical Layer is an exception**

The Open System Interconnection (OSI) Model

- International Organization for Standardization (ISO)
- ISO established in 1947
- **Close to three-fourths of countries represented**
- Introduced OSI Model in late 1970s
- **OSI: a 7-Layer Model**

OSI Model vs TCP/IP Protocol suite

- Two Layers of OSI missing from TCP/IP

- **Application (TCP/IP) = Application + Presentation + Session (OSI)**

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Encapsulation and Decapsulation

One of the important concepts in protocol layering in the Internet is encapsulation/Decapsulation

OSI versus TCP/IP

When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite. These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model. The application layer in the suite is usually considered to be the combination of three layers in the OSI model. Two reasons were mentioned for this decision. First, TCP/IP has more than one 31

transport-layer protocol. Some of the functionalities of the session layer are available in some of the transport-layer protocols. Second, the application layer is not only one piece of software. Many applications can be developed at this layer. If some of the functionalities mentioned in the session and presentation layers are needed for a particular application, they can be included in the development of that piece of software.

Lack of OSI Model's Success

The OSI model appeared after the TCP/IP protocol suite. Most experts were at first excited and thought that the TCP/IP protocol would be fully replaced by the OSI model. This did not happen for several reasons, but we describe only three, which are agreed upon by all experts in the field. First, OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot. Second, some layers in the OSI model were never fully defined. For example, although the services provided by the presentation and the session layers 32

were listed in the document, actual protocols for these two layers were not fully defined, nor were they fully described, and the corresponding software was not fully developed. Third, when OSI was implemented by an organization in a different application, it did not show a high enough level of performance to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.

Summary

A protocol is a set of rules that governs communication. In protocol layering, we need to follow two principles to provide bidirectional communication. First, each layer needs to perform two opposite tasks. Second, two objects under each layer at both sides should be identical. In a protocol layering, we need to distinguish between a logical connection and a physical connection. Two protocols at the same layer can have a logic call

connection; a physical connection is only possible through the physical layers.

TCP/IP is a hierarchical protocol suite made of five layers: physical, data link, network, transport, and application. The physical layer coordinates the functions required

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to transmit a bit stream over a physical medium. The data-link layer is responsible for delivering data units from one station to the next without errors. The network layer is responsible for the source-to-destination delivery of a packet across multiple network links. The transport layer is responsible for the process-to-process delivery of the entire message. The application layer enables the users to access the network.

Four levels of addresses are used in an internet following the TCP/IP protocols: physical (link) addresses, logical (IP) addresses, port addresses, and specific addresses. The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN. The IP address uniquely defines a host on the Internet. The port address identifies a process on a host. A specific address is a user-friendly address.

Another model that defines protocol layering is the Open Systems Interconnection (OSI) model. Two layers in the OSI model, session and presentation, are missing from the TCP/IP protocol suite. These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model. The application layer in the suite is usually considered to be the combination of three layers in the OSI model. The OSI model did not replace the TCP/IP protocol suite because it was completed when TCP/IP was fully 34

in place and because some layers in the OSI model were never fully defined.

CHP#03

Communication at Physical Layer

Analog & digital data:

Analog and Digital Data

Data can be analog or digital. The term analog data refers to information that is 35

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continuous; digital data refers to information that has discrete states. For example, an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous.

Analog and Digital Signals

Like the data they represent, signals can be either analog or digital. An analog signal has infinitely many levels of intensity over a period of time. As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path. A digital signal, on the other hand, can have only a limited number of defined 36

values. Although each value can be any number, it is often as simple as 1 and 0. The simplest way to show signals is by plotting them on a pair of perpendicular axes. The vertical axis represents the value or strength of a signal. The horizontal axis represents time. Figure 3.2 illustrates an analog signal and a digital signal. The curve representing the analog signal passes through an infinite number of points. The vertical lines of the digital signal, however, demonstrate the sudden jump that the signal makes from value to value. 37

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Periodic & Non-periodic Signals

Both analog and digital signals can take one of two forms: periodic or nonperiodic (sometimes referred to as aperiodic; the prefix a in Greek means “non”).

A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a cycle. A nonperiodic signal changes without exhibiting a pattern

or cycle that repeats over time. 38

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Both analog and digital signals can be periodic or nonperiodic. In data communications, we commonly use periodic analog signals and nonperiodic digital signals

In data communications, we commonly use periodic analog signals and nonperiodic digital signals

PERIODIC ANALOG SIGNALS

- **Periodic Analog Signals → Simple or Composite**
- **Simple Periodic Analog signal → Sine wave**
- **Composite Periodic Analog signal → Composed of multiple sine waves**

Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

Sine Wave

The sine wave is the most fundamental form of a periodic analog signal. When we visualize it as a simple oscillating curve, its change over the course of a cycle is smooth and consistent, a continuous, rolling flow.

A sine wave can be represented by three parameters: the peak amplitude, the frequency, and the phase. These three parameters fully describe a sine wave

Peak Amplitude

The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in volts.

Period and Frequency

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

Frequency refers to the number of periods in 1 s. Note that period and frequency are just one characteristic defined in two ways. Period is the inverse of frequency, and frequency is the inverse of period, as the following formulas show. 40 41

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More About Frequency

We already know that frequency is the relationship of a signal to time and that the frequency of a wave is the number of cycles it completes in 1 s. But another way to look at frequency is as a measurement of the rate of change. Electromagnetic signals are oscillating waveforms; that is, they fluctuate continuously and predictably above and below a mean energy level. A 40-Hz signal has one-half the frequency of an 80-Hz signal; it completes 1 cycle in twice the time of the 80-Hz signal, so each cycle also takes twice as long to change from its lowest to its highest voltage levels. Frequency, therefore, though described in cycles per second

(hertz), is a general measurement of the rate of change of a signal with respect to time.

Frequency is the rate of change with respect to time. Change in a short span of time means high frequency. Change over a long span of time means low frequency.

- If the value of a signal changes over a very short span of time, its frequency is high.
- If it changes over a long span of time, its frequency is low.
- If a signal does not change at all, its frequency is zero.
- If a signal changes instantaneously, its frequency is infinite

Phase describes the position of the waveform relative to time 0

2π

$2\pi/$

Phase is measured in degrees or radians [360° is

rad; 1° is

360 rad , and 1 rad

2π].

is $360/(\pi)$

A phase shift of 360° corresponds to a shift of a complete period; a phase

180°

shift of

corresponds to a shift of one-half of a period; and a phase shift of 90° corresponds to a shift of one-quarter of a period 43

A sine wave is offset $\frac{1}{6}$ cycle with respect to time 0. What is its phase in degrees and radians? 44

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0°

a. A

sine wave with a phase of
starts at time 0 with a zero amplitude. The
amplitude is increasing.

90°

b. A

sine wave with a phase of
starts at time 0 with a peak amplitude. The
amplitude is decreasing.

180°

c. A

sine wave with a phase of
starts at time 0 with a zero amplitude. The
amplitude is decreasing.

Another way to look at the phase is in terms of shift or offset. We can say that

0°

a. A

sine wave with a phase of
is not shifted.

90°

b. A 45

sine wave with a phase of
is shifted to the left by
cycle. However, note
that the signal does not really exist before time 0.
 180°

c. A
sine wave with a phase of
is shifted to the left by
cycle. However, note
that the signal does not really exist before time 0.

Wavelength

Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium

CS601 UPDATE NOTES

BY SONUMUGHAL

9.4.2 Questions

Q9-1. Distinguish between communication at the network layer and communication at the data-link layer.

Ans: Communication at the network layer is host-to-host, which is responsible for end to end packet delivery and packet routing through intermediate hosts.

Communication at the data-link layer is node-to-node on the same local network.

Q9-2. Distinguish between a point-to-point link and a broadcast link.

Ans: A broadcast link is the link that connects two or more nodes where one node can transmit the signal for the rest of the nodes to receive the signal at the same time. Point to point link is the link that connects just two nodes or endpoints where one node sends the signal and the other node receives it.

Q9-3. Can two hosts in two different networks have the same link-layer address? Explain.

Ans: Two hosts in two different networks can theoretically have the same link-layer address because a link layer address has only local jurisdiction.

Q9-4. Is the size of the ARP packet fixed? Explain.

Ans: No, it can have two MAC addresses in it and two different protocol addresses in it. The size depends upon the network and data-link protocol.

Q9-5. What is the size of an ARP packet when the protocol is IPv4 and the hardware is Ethernet?

Ans: 28bytes = 2+2+1+1+2+6+4+6+4

Q9-6. Assume we have an isolated link (not connected to any other link) such as a private network in a company. Do we still need addresses in both the network layer and the data-link layer? Explain.

Ans: each hop (router or host) should know its own link-layer address. The destination link-layer address is determined by using the address Resolution protocol.

Q9-7. In Figure 9.9, why is the destination hardware address all 0s in the ARP request message?

Ans: The frame uses a broadcast address so that the intended recipient on the network will receive the frame even though the sender does not know the specific hardware address of the intended recipient.

Q9-8. In Figure 9.9, why is the destination hardware address of the frame from A to B a broadcast address?

Q9-9. In Figure 9.9, how does system A know what the link-layer address of system B is when it receives the ARP reply?

Q9-10. When we talk about the broadcast address in a link, do we mean sending a message to all hosts and routers in the link or to all hosts and routers in the Internet? In other words, does a broadcast address have a local jurisdiction or a universal jurisdiction? Explain.

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Q9-11. Why does a host or a router need to run the ARP program all of the time in the background?

Q9-12. Why does a router normally have more than one interface?

Q9-13. Why is it better not to change an end-to-end address from the source to the destination?

Q9-14. How many IP addresses and how many link-layer addresses should a router have when it is connected to five links?

9.4.3 Problems

P9-1. Assume we have an internet (a private small internet) in which all hosts are connected in a mesh topology. Do we need routers in this internet? Explain.

P9-2. In the previous problem, do we need both network and data-link layers?

P9-3. Explain why we do not need the router in Figure 9.15.

P9-4. Explain why we may need a router in Figure 9.16.

P9-5. Is the current Internet using circuit-switching or packet-switching at the datalink layer? Explain.

P9-6. Assume Alice is travelling from 2020 Main Street in Los Angeles to 1432 American Boulevard in Chicago. If she is travelling by air from Los Angeles Airport to Chicago Airport,

a. find the end-to-end addresses in this scenario.

b. find the link-layer addresses in this scenario.

P9-7. In the previous problem, assume Alice cannot find a direct flight from the Los Angeles to the Chicago. If she needs to change flights in Denver,

a. find the end-to-end addresses in this scenario.

b. find the link-layer addresses in this scenario.

P9-8. When we send a letter using the services provided by the post office, do we use an end-to-end address? Does the post office necessarily use an end-to-end address to deliver the mail? Explain.

Figure 9.15 Problem 9-3

Figure 9.16 Problem 9-4

Alice's site

Alice Bob

Bob's site

R

CHAPTER 9 INTRODUCTION TO DATA-LINK LAYER 255

P9-9. In Figure 9.5, assume Link 2 is broken. How can Alice communicate with Bob?

P9-10. In Figure 9.5, show the process of frame change in routers R1 and R2.

P9-11. In Figure 9.7, assume system B is not running the ARP program. What would happen?

P9-12. In Figure 9.7, do you think that system A should first check its cache for mapping from N2 to L2 before even broadcasting the ARP request?

P9-13. Assume the network in Figure 9.7 does not support broadcasting. What do you suggest for sending the ARP request in this network?

P9-14. In Figures 9.11 to 9.13, both the forwarding table and ARP are doing a kind of mapping. Show the difference between them by listing the input and output of mapping for a forwarding table and ARP.

P9-15. Figure 9.7 shows a system as either a host or a router. What would be the

actual entity (host or router) of system A and B in each of the following cases:

- a. If the link is the first one in the path?
- b. If the link is the middle one in the path?
- c. If the link is the last one in the path?
- d. If there is only one link in the path (local communication)?

Question: Give some detail about "fault tolerance".

Answer: Fault tolerance means the system will not fail because any one component fails. The system also should provide recovery from multiple failures. Components are often over engineered or purposely underutilized to ensure that while performance may be affected during an outage the system will perform within predictable, acceptable bounds.

Question: What are regulatory Agencies?

Answer: All communication technology is subject to regulation and laws by government agencies. The purpose is to protect Public Interest by regulating Radio, Television and Cable Communications.

Question: What is the main problem with ASK (amplitude Shift Keying)?

Answer: The main problem with ASK is the noise. Noise usually affected the amplitude so it is most affected by Noise.

Question: What are the basic components of a data communication system ?

Answer: There are five basic components of a data communication. These are as follows:- 1) message 2) sender 3) receiver 4) medium 5) protocol

Question: How can I define Data communication?

Answer: Data communication is the name of a communication between computers. More specifically, the exchange of data in the form 0's and 1's between two devices (computers) via some form of the transmission medium. Nowadays, data communication between computers also supports digital speech telephone and videophone, thus following direct communication between people.

Question: What is PCMCIA?

Answer: PCMCIA (Personal Computer Memory Card International Association) is an international standards body and trade association with over 100 member companies that was founded in 1989 to establish standards for Integrated Circuit cards and to promote interchangeability among mobile computers where ruggedness, low power, and small size were critical. As the needs of mobile computer users has changed, so has PCMCIA. By 1991, PCMCIA had defined an I/O interface for the same 68 pin connector initially used for memory cards. At the same time, the Socket Services Specification was added and was soon followed by the Card Services Specification as developers realized that common software would be needed to enhance compatibility.

Question: Will VoIP work with DSL?

Answer: Yes, VoIP works with DSL. Many of the phone companies currently providing DSL and landline phone services are in the process of unbundling DSL and phone services.

Question: What is attenuation?

Answer: Attenuation is signal loss due to the diminishing availability of signal energy, or signal power. As a analog or digital signal traverses across a medium, it fades. High attenuation may lead to the inability to recover the signal on the far end. Signal repeaters may be used on the transmission path to periodically boost the signal strength. Baseband transmission is extremely limited to attenuation. Broad-band much less so, In addition, wireless communications is much less susceptible to attenuation that is wire-line communications such as x-DSL or cable modems.

Question: What is difference between logical address and physical address?

Answer: The OSI model is a good place to start to learn more about the differences between physical and logical addressing. Think of the physical address as the 48-bit MAC address that manufacturers encode in their network interface cards (NICs). This type of address is unique, referred to as the Ethernet or hardware address, and cannot be changed. The MAC or Ethernet address is associated with Layer 2 (data Link) of the OSI Model. The logical address is a 32-bit IP address that is not embedded in the network card but it is assigned to it

for the purpose of routing between networks. This type of address operates at Layer 3 (network) of the OSI Model.

Question: What is the differences between OSI and TCP/IP Model?

Answer: The ISO/OSI protocol is a 7 layer reference model that was designed after TCP/IP model, which only has 4 layers. That said, the differences between the two are only minor.

The 7 layers of the OSI Protocol Stack are as follows:

7. Application: end user services such as email
6. Presentation: data problems and data compression
5. Session: authentication and authorization
4. Transport: guarantee end-to-end delivery of packets
3. Network: packet routing
2. Data Link: transmit and receive packets
1. Physical: The physical connection or cable itself.

Of these 7 layers, the 4 layers the TCP/IP Protocol Stack use are as follows:

5. Application: authentication, compression, and end user services
4. Transport: handles the flow of data between systems and provides access to the network for applications via the BSD socket library
3. Network: packet routing
2. Link: Kernel OS/device driver interface to the network interface on the computer.

The main differences between OSI and TCP/IP are:

The application layer in TCP/IP handles the responsibilities of layers 5,6, and 7 in the OSI model. The transport layer in TCP/IP does not always guarantee reliable delivery of packets as the transport layer in the OSI model does.

So as you can see there is not too much difference between the models. In terms of software implementation TCP/IP looks like this:

Application Layer: some of the applications covered are SMTP (mail), Telnet, FTP, Rlogin, NFS, NIS, and LPD.

Transport Layer: the transport uses two protocols, UDP (User Datagram Protocol) and TCP, which does not guarantee packet delivery and applications which use this must provide their own means of verifying delivery. TCP guarantees delivery of packets to the applications that use it.

Network Layer: the network layer is concerned with packet routing and used low level protocols such as ICMP, IP, and IGMP.

Link Layer: the link layer is concerned with the actual transmittal of packets as well as IP to Ethernet address translation. This layer is concerned with Arp, the device driver and Rarp.

What is Tx and Rx?

Answer: Tx is the abbreviation used for transmission/ transmitter and Rx is receiver.

Question: What do we mean by wireless networking?

Answer: The term wireless networking refers to technology that enables two or more computers to communicate using standard network protocols, but without network cabling. Strictly speaking, any technology that does this could be called wireless networking. The current buzzword however generally refers to wireless LANs.

Question: What is PDU in OSI Model?

Answer: Normally a communication request originates at the highest layer (Application Layer). The request is passed down through the lower layers in the form of a packet called a protocol data unit (PDU). Layers in the protocol stack communicate with their adjacent layers via one or more Service Access Points (SAP). Each succeeding layer in the stack adds its own information to the PDU

that will be read by its counterpart (peer) layer on the receiving system. Once the data arrives at the lower layers, the PDU is encoded into data frames and placed onto the cable for transmission. The data frames make their way to the receiving system and the entire process is reversed as the PDU makes its way up the protocol stack. As it moves up the stack, each layer "unwrap" the PDU and receives the information from its peer layer on the sending system.

Question: What is the difference between WAP and GPRS?

Answer: Wireless Application Protocol, a secure specification that allows users to access information instantly via handheld wireless devices such as mobile phones. Wireless Application Protocol (or WAP) is envisioned as a comprehensive and scalable protocol designed to use with mobile phones using Short Message Services (SMS), General Packet Radio Services (GPRS), CDMA and GSM.

Question: What are routers, repeaters and bridges?

Answer: Router: Routers relay packets among multiple interconnection networks. Router is device that determines the next network point to which a data packet should be forwarded enroute toward its destination. Routers create or maintain a table of the available routes and use this information to determine the best route for a given data packet.

Question: What do we mean by capacity of the channel?

Answer: Channel capacity: The maximum possible information transfer rate through a channel, subject to specified constraints.

Question: What is remote data communication?

Answer: Data Communication is considered remote, if the devices are farther apart.

Question: Explain VPN (Virtual Private Network)

Answer: A virtual private network (VPN) is a way to use a public telecommunication infrastructure, such as the Internet, to provide remote offices or individual users with secure access to their organization's network. A virtual private network can be contrasted with an expensive system of owned or leased

lines that can only be used by one organization. The goal of a VPN is to provide the organization with the same capabilities, but at a much lower cost.

Question: Define Protocols

Answer: Protocol is an agreed-upon format for transmitting data between two devices. The protocol determines the following: 1) The type of error checking to be used. 2) Data compression method, if any. 3) How the sending device will indicate that it has finished sending a message? 4) How the receiving device will indicate that it has received a message?

Question: Define encryption of data.

Answer: Encryption of data: The translation of data into a secret code. Encryption is the most effective way to achieve data security. To read an encrypted file, you must have access to a secret key or password that enables you to decrypt it. Unencrypted data is called plain text; encrypted data is referred to as cipher text.

Can we implement OSI in wireless networks?

Answer: Wireless Application Protocol (WAP), a secure specification that allows users to access information via handheld wireless devices, specifies architecture based on layers that follow the OSI model fairly closely. WAP defines network architecture for content delivery over wireless networks. Central to the design of WAP is a network stack based on the OSI model. WAP implements several new networking protocols that perform functions similar to the well-known Web protocols HTTP, TCP, and SSL.

Question: What is distributed processing?

Answer: It refers to any of a variety of computer systems that use more than one computer, or processor, to run an application. This includes parallel processing, in which a single computer uses more than one CPU to execute programs. More often, however, distributed processing refers to local-area networks (LANs) designed so that a single program can run simultaneously at various sites. Most distributed processing systems contain sophisticated software that detects idle CPUs on the network and parcels out programs to utilize them. Another form of distributed processing involves distributed databases, databases in

which the data is stored across two or more computer systems. The database system keeps track of where the data is so that the distributed nature of the database is not apparent to users.

Question: What is a sample?

Answer: Sample is the amplitude of an analog signal at some specific interval. It is achieved in pulse amplitude modulation, which is the first step towards analog to digital conversion.

Question: What Is Line Discipline?

Answer: Whatever the system, no device in it should be allowed to transmit until that device has evidence that the intended receiver is able to receive and is prepared to accept the transmission. What if the receiver does not expect a transmission, is busy, or is out of commission? With no way to determine the status of intended receiver, the transmitting device may waste its time sending data to a nonfunctioning receiver or may interfere with signals already on the link. The line discipline function of the data link layer oversees the establishment of links and the right of a particular device to transmit at a given time. Line discipline answers the question. Who should send now?

Question: Is it possible for digital data to be transmitted via ordinary phone line?

Answer: The Digital Subscriber line (DSL) is a newer technology that uses the existing telecommunication networks such as the local loop telephone line (still an analog line) to accomplish high speed delivery of data, voice, video and multimedia.

Question: What is then difference between T-lines and E-Lines?

Answer: Europeans use a version of T lines called E lines. The two systems are conceptually identical, but their capacities differ.

Question: What do we mean by the term session?

Answer: Session is the period of time a user interfaces with an application. The user session begins when the user accesses the application and ends when the user

quits the application.

Question: How can I define Data communication?

Answer: Data Communication is the exchange of data (in the form of 0's and 1's) between two devices (computers) via some form of the transmission medium. Data communication is communication between computers. Data communication concerns the exchange of digital data between computers. Nowadays, data communication between computers also supports digital speech telephone and videophone, thus supporting direct communication between people.

Question: How could it be more secure using distributed processing?

Answer: In this case security is achieved by providing the user with limited access. It means that user can only perform those tasks which are allowed by system designer. Its' simple example is bank's ATM. Where user can't perform operations on database other than provided.

Question: What does the terms transit and response time mean? Also tell me about the topic peak load periods.

Answer: Transit time is the time taken by the message to travel from source to destination. Response time is the time of response (acknowledgement) from destination to source. Peak load periods are those periods of time in which number of users on a network are more than normal time.

What are the advantages of networking?

Answer: • Files can be stored on a central computer (the file server) allowing data to be shared throughout an organization. • Files can be backed up more easily when they are all on a central fileserver rather than when they are scattered across a number of independent workstations. • Networks also allow security to be established, ensuring that the network users may only have access to certain files and applications. • Software and resources can be centrally managed. • Network versions of software often allow for their speedy installation on workstations from the file server. • Expensive devices such as laser printers or scanners can be shared. • Users can access their files from any workstation

Question: What are different factors to choose a network topology?

Answer: Working conditions of network, working scenarios, arrangement of links and devices. Depending upon our network what actually we are dealing with and in which geographical area our devices are present, small area available or have large area. All these factors are considered when choosing a Topology.

Question: What is the relative status of the devices to be linked?

Answer: The geographical area in which our devices are present is the relative status of devices.

Question: What is the definition of Hub?

Answer: Hub is a common connection point for devices in a network. Hubs are commonly used to connect segments of a LAN. A hub contains multiple ports. When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets. A passive hub serves simply as a conduit for the data, enabling it to go from one device (or segment) to another. So-called intelligent hubs include additional features that enable an administrator to monitor the traffic passing through the hub and to configure each port in the hub. Intelligent hubs are also called manageable hubs. A third type of hub, called a switching hub, actually reads the destination address of each packet and then forwards the packet to the correct port.

Question: What do we mean by "TAP"?

Answer: A tap is a connection to a coaxial cable in which a hole is drilled through the outer shield of the cable so that a clamp can be connected to the inner conductor of the cable. Instead of cutting the cable and attaching connectors to both ends of the severed coaxial cable,

Question: Give some detail about "fault tolerance".

Answer: Fault tolerance means the system will not fail because any one component fails. The system also should provide recovery from multiple failures. Components are often over engineered or purposely underutilized to ensure that while performance may be affected during an outage the system will perform within predictable, acceptable bounds

Question: What is the difference between radio and microwave?

Answer: Although there is no clear-cut demarcation between radio and microwave, electromagnetic waves ranging in frequencies between 3 KHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 GHz and 300 GHz are called microwaves. Radio waves are used for multicast communications, such as radio and television systems.

Question: What is the difference between internet and intranet?

Answer: Internet The Internet is a system of linked networks that are worldwide in scope and facilitate data communication services such as remote login, file transfer, electronic mail, the World Wide Web and newsgroups. With the meteoric rise in demand for connectivity, the Internet has become a communications highway for millions of users. The Internet was initially restricted to military and academic institutions, but now it is a full-fledged conduit for any and all forms of information and commerce. Internet websites now provide personal, educational, political and economic resources to every corner of the planet. Intranet A network based on TCP/IP protocols (an internet) belonging to an organization, usually a corporation, accessible only by the organization's members, employees, or others with authorization. An intranet's Web sites look and act just like any other Web sites, but the firewall surrounding an intranet fends off unauthorized access. Like the Internet itself, intranets are used to share information. Secure intranets are now the fastest-growing segment of the Internet because they are much less expensive to build and manage than private networks based on proprietary protocols.

Question: What is the difference between hub and switch?

Answer: On the outside, hubs and switches appear very similar in that they both have a number of RJ-45 jacks for connecting devices. Inside, however, they work very differently. To understand why switches provide so much more functionality than hubs, you must understand a fundamental limitation of (non-switched) Ethernet: there can only be one device transmitting on a segment at any given time. If two or more devices attempt to transmit at the same time, a collision occurs. (In fact, an Ethernet segment where only one conversation can occur is called a collision domain.) After a collision, all devices must retransmit. As you can imagine, as the number of devices on an Ethernet segment increases,

the probability for collisions increase. Because devices must spend more time retransmitting data, the network is perceived to be slow. Before the advent of switches, a network could be divided into segments with a device called a bridge. Bridges have two Ethernet ports. As traffic flows through a network, a bridge learns which devices (identified by the MAC or "hardware" address) are on each side. The bridge then makes decisions to forward or not forward each packet to the other side based on where the destination device is located. A bridge thus divides a network into two collision domains, allowing two independent "conversations" to occur. If a bridge is placed intelligently (e.g., separating two departments and their respective file servers), they can improve network efficiency. Hubs do no processing on network traffic--they simply repeat the incoming signal to all available ports. On a switch, every port acts as a bridge. If each switch port is connected to a single device, each device can, in principle, act independently of every other device. For example, consider a switch with the following devices attached: computer 1 computer 2 computer 3 printer file server uplink to the Internet In this case, computer 1 could be printing a document, while computer 2 connects to a files server, while computer 3 accesses the Internet. Because the switch intelligently forwards traffic only to the devices involved, there can be multiple independent simultaneous conversations.

Question: How different layers of OSI model are implemented?

Answer: The OSI 7 layers model has clear characteristics. Layers 7 through 4 deal with end to end communications between data source and destinations. Layers 3 to 1 deal with communications between network devices. On the other hand, the seven layers of the OSI model can be divided into two groups: upper layers (layers 7, 6 & 5) and lower layers (layers 4, 3, 2, 1). The upper layers of the OSI model deal with application issues and generally are implemented only in software. The highest layer, the application layer, is closest to the end user. The lower layers of the OSI model handle data transport issues. The physical layer and the data link layer are implemented in hardware and software. The lowest layer, the physical layer, is closest to the physical network medium (the wires, for example) and is responsible for placing data on the medium.

How can we define networking?

Answer: In information technology, networking is the construction, design, and use of network, including the physical (cabling, hub, bridge, switch, router, and so forth), the selection and use of telecommunication protocol and computer software for using and managing the network, and the establishment of operation policies and procedures related to the network

Question: What is walkie talkie?

Answer: A handie talkie, often referred to by its abbreviation, HT, is a handheld, portable two-way radio transceiver. This type of radio is sometimes called a "walkie talkie" or a "handheld." Handie talkies are popular among amateur radio operators, especially on their VHF and UHF bands at 144 and 432 MHz. Handie talkies are widely used by security personnel, military personnel, and police officers. Most HTs are used in conjunction with repeaters for extended range. Some HTs are designed for the 27-MHz Citizens Band (CB) radio service. A typical HT is a rectangular box about the size and weight of an old-fashioned telephone handset. The antenna protrudes from the top end, and consists of a coiled-up element encased in rubber and wound around a flexible rod. This type of antenna, known as a "rubber duck," is not particularly efficient, but is convenient and rugged. Volume and squelch controls are usually placed next to the antenna. The frequency control knob or buttons are on the top end or the front.

KINDLY PRAY FOR OUR COUNTRY AND SOLDIERS

BY SONU MUGHAL

27/2/19

13.2

IEEE Project 802

-In 1985, the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.

-It is a way of specifying functions of the physical layer and the data-link layer of major LAN protocols.

9.4.2 Questions

Q9-1. Distinguish between communication at the network layer and communication at the data-link layer.

Ans: Communication at the network layer is host-to-host, which is responsible for end to end packet delivery and packet routing through intermediate hosts.

Communication at the data-link layer is node-to-node on the same local network.

Q9-2. Distinguish between a point-to-point link and a broadcast link.

Ans: A broadcast link is the link that connects two or more nodes where one node can transmit the signal for the rest of the nodes to receive the signal at the same time. Point to point link is the link that connects just two nodes or endpoints where one node sends the signal and the other node receives it.

Q9-3. Can two hosts in two different networks have the same link-layer address? Explain.

Ans: Two hosts in two different networks can theoretically have the same link-layer address because a link layer address has only local jurisdiction.

Q9-4. Is the size of the ARP packet fixed? Explain.

Ans: No, it can have two MAC addresses in it and two different protocol addresses in it. The size depends upon the network and data-link protocol.

Q9-5. What is the size of an ARP packet when the protocol is IPv4 and the hardware is Ethernet?

Ans: 28bytes = 2+2+1+1+2+6+4+6+4

Q9-6. Assume we have an isolated link (not connected to any other link) such as a private network in a company. Do we still need addresses in both the network layer and the data-link layer? Explain.

Ans: each hop (router or host) should know its own link-layer address. The destination link-layer address is determined by using the address Resolution protocol.

Q9-7. In Figure 9.9, why is the destination hardware address all 0s in the ARP request message?

Ans: The frame uses a broadcast address so that the intended recipient on the network will receive the frame even though the sender does not know the specific hardware address of the intended recipient.

Q9-8. In Figure 9.9, why is the destination hardware address of the frame from A to B a broadcast address?

Q9-9. In Figure 9.9, how does system A know what the link-layer address of system B is when it receives the ARP reply?

Q9-10. When we talk about the broadcast address in a link, do we mean sending a message to all hosts and routers in the link or to all hosts and routers in the Internet? In other words, does a broadcast address have a local jurisdiction or a universal jurisdiction? Explain.

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Q9-11. Why does a host or a router need to run the ARP program all of the time in the background?

Q9-12. Why does a router normally have more than one interface?

Q9-13. Why is it better not to change an end-to-end address from the source to the

destination?

Q9-14. How many IP addresses and how many link-layer addresses should a router have when it is connected to five links?

9.4.3 Problems

P9-1. Assume we have an internet (a private small internet) in which all hosts are connected in a mesh topology. Do we need routers in this internet? Explain.

P9-2. In the previous problem, do we need both network and data-link layers?

P9-3. Explain why we do not need the router in Figure 9.15.

P9-4. Explain why we may need a router in Figure 9.16.

P9-5. Is the current Internet using circuit-switching or packet-switching at the datalink layer? Explain.

P9-6. Assume Alice is travelling from 2020 Main Street in Los Angeles to 1432 American Boulevard in Chicago. If she is travelling by air from Los Angeles Airport to Chicago Airport,

a. find the end-to-end addresses in this scenario.

b. find the link-layer addresses in this scenario.

P9-7. In the previous problem, assume Alice cannot find a direct flight from the Los Angeles to the Chicago. If she needs to change flights in Denver,

a. find the end-to-end addresses in this scenario.

b. find the link-layer addresses in this scenario.

P9-8. When we send a letter using the services provided by the post office, do we use an end-to-end address? Does the post office necessarily use an end-to-end address to deliver the mail? Explain.

Figure 9.15 Problem 9-3

Figure 9.16 Problem 9-4

Alice's site

Alice Bob

Bob's site

R

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P9-9. In Figure 9.5, assume Link 2 is broken. How can Alice communicate with Bob?

P9-10. In Figure 9.5, show the process of frame change in routers R1 and R2.

P9-11. In Figure 9.7, assume system B is not running the ARP program. What would happen?

P9-12. In Figure 9.7, do you think that system A should first check its cache for mapping from N2 to L2 before even broadcasting the ARP request?

P9-13. Assume the network in Figure 9.7 does not support broadcasting. What do you suggest for sending the ARP request in this network?

P9-14. In Figures 9.11 to 9.13, both the forwarding table and ARP are doing a kind of mapping. Show the difference between them by listing the input and output of mapping for a forwarding table and ARP.

P9-15. Figure 9.7 shows a system as either a host or a router. What would be the

actual entity (host or router) of system A and B in each of the following cases:

- a.** If the link is the first one in the path?
- b.** If the link is the middle one in the path?
- c.** If the link is the last one in the path?
- d.** If there is only one link in the path (local communication)?

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